

Dynamical Processes for Descriptive Ocean Circulation

The complete version of this chapter (Chapter S7) appears on the textbook Web site <http://booksite.academicpress.com/DPO/>. The sections and equations are identical, but the explanatory text and figures are greatly truncated in this book. Figures, chapters, and sections that appear only on the Web site are denoted by “S” in their name such as Figure S7, Chapter S7, and so forth. Tables mentioned in this chapter appear only on the Web site.

7.1. INTRODUCTION: MECHANISMS

Ultimately, motion of water in the ocean is driven by the sun, the moon, or tectonic processes. The sun’s energy is transferred to the ocean through buoyancy fluxes (heat fluxes and water vapor fluxes) and through the winds. Tides create internal waves that break, creating turbulence and mixing. Earthquakes and turbidity currents create random, irregular waves including tsunamis. Geothermal processes heat the water very gradually with little effect on circulation.

Earth’s rotation profoundly affects almost all phenomena described in this text. Rotating fluids behave differently from non-rotating fluids in ways that might be counterintuitive.

In a non-rotating fluid, a pressure difference between two points in the fluid drives the fluid toward the low pressure. In a fluid dominated by rotation, the flow can be *geostrophic*, perpendicular to the pressure gradient force, circling around centers of high or low pressure due to the Coriolis effect.

Ocean circulation is often divided conceptually into *wind-driven* and *thermohaline* (or buoyancy-dominated) components. Wind causes waves, inertial currents, and Langmuir cells. At longer timescales, which involve the Coriolis effect, wind drives the near-surface frictional layer and, indirectly, the large-scale gyres and currents that are usually referred to as the wind-driven circulation. Thermohaline circulation is associated with heating and cooling (“thermo”), and evaporation, precipitation, runoff, and sea ice formation, all of which change salinity (“haline”). Thermohaline-dominated circulation is mostly weak and slow compared with wind-driven circulation. In discussing thermohaline effects, it is common to refer to the *overturning circulation*, which involves buoyancy changes. The energy source for thermohaline circulation importantly includes the wind and tides that produce the turbulence that is essential for the diffusive upwelling across isopycnals that closes the thermohaline overturning. Both the wind-driven

and thermohaline circulations are almost completely in geostrophic balance, with the forcing that drives them occurring at higher order.

7.2. MOMENTUM BALANCE

Fluid flow in three dimensions is governed by three equations expressing how velocity (or momentum) changes, one for each of the three physical dimensions. Each of the three momentum equations includes an acceleration term (how velocity changes with time), an advection term (see Section 5.1.3), and forcing terms:

$$\begin{aligned} \text{Density} \times (\text{Acceleration} + \text{Advection}) \\ = \text{Forces per unit volume} \end{aligned} \quad (7.1)$$

$$\begin{aligned} \text{Forces per unit volume} = \text{Pressure} \\ \text{gradient force} + \text{Gravity} + \text{Friction} \end{aligned} \quad (7.2)$$

Expressions (7.1) and (7.2) are each three equations, one for each of the three directions (e.g., east, north, and up). The terms in Eqs. (7.1) and (7.2) are illustrated in Figure 7.1.

The inclusion of advection means that Eq. (7.1) is the expression of momentum change in a *Eulerian* framework, where the observer sits at a fixed location relative to Earth. Equation (7.1) can be written without the advection term, in a *Lagrangian* framework, where the observer drifts along with the fluid flow. (See also Section S16.5 in the online Supplement.)

For a rotating geophysical flow, we, as observers, sit within a rotating “frame of reference” attached to the rotating Earth. For this reference frame, the acceleration term on the left-hand side of Eq. (7.1) is rewritten to separate local acceleration due to an actual local force from the effects of rotation. The effects that are separated out are the *centrifugal* and *Coriolis accelerations* (Section 7.2.3).

The frictional force in Eq. (7.1) leads to *dissipation* of energy due to the fluid’s *viscosity*.

7.2.1. Acceleration and Advection

Acceleration is the change in velocity with time. If the vector velocity is expressed in Cartesian coordinates as $\mathbf{u} = (u, v, w)$ where the bold $\mathbf{u}$ indicates a vector quantity, and u , v , and w are the positive eastward (x-direction), northward (y-direction) and positive upward (z-direction) velocities, then

$$\text{x-direction acceleration} = \partial u / \partial t \quad (7.3a)$$

with similar expressions for the y - and z -directions.

Advection is defined in Section 5.1.3. Advection is how the flow moves properties (including scalars such as temperature or salinity) and vectors (such as the velocity). Advection can change the flow property if there is a gradient in the property through which the fluid moves. In the x-momentum equation, the advection term is

$$\begin{aligned} \text{x-direction advection} \\ = u \partial u / \partial x + v \partial u / \partial y + w \partial u / \partial z \end{aligned} \quad (7.3b)$$

The *substantial derivative* is the sum of the acceleration and advection terms:

$$\begin{aligned} Du/Dt = \partial u / \partial t + u \partial u / \partial x + v \partial u / \partial y \\ + w \partial u / \partial z \end{aligned} \quad (7.4)$$

7.2.2. Pressure Gradient Force and Gravitational Force

Pressure is defined in Section 3.2. The flow of fluid due to spatial variations in pressure is also described. In mathematical form, the pressure gradient force is

$$\begin{aligned} \text{x-direction pressure gradient force} \\ = -\partial p / \partial x \end{aligned} \quad (7.5)$$

The *gravitational force* between Earth and the object or fluid parcel is directed toward the center of mass of Earth. Gravitational force is

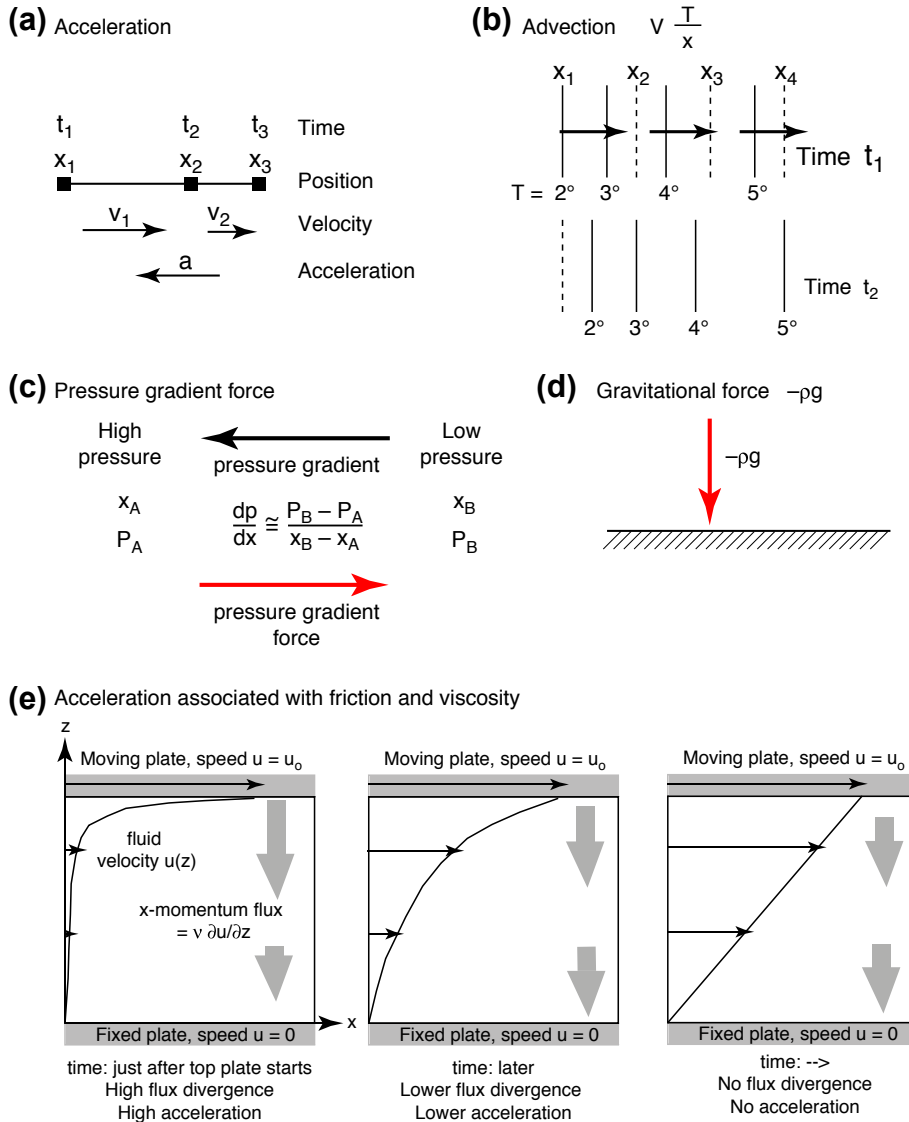


FIGURE 7.1 Forces and accelerations in a fluid: (a) acceleration, (b) advection, (c) pressure gradient force, (d) gravity, and (e) acceleration associated with viscosity ν .

mass of the object $\times$ gravitational acceleration g , equal to $9.780318 \text{ m}^2/\text{sec}$ (at the equator). The gravitational force per unit volume is

$$\begin{aligned} \text{z-direction gravitational force per unit} \\ \text{volume} = -\rho g \end{aligned} \quad (7.6)$$

7.2.3. Rotation: Centrifugal and Coriolis Forces

Centrifugal force is the apparent outward force on a mass when it is rotated. Since Earth rotates around a fixed axis, the direction of centrifugal force is always outward away from the axis,

opposite to the direction of gravity at the equator; at Earth's poles it is zero. (*Centripetal force* is the necessary inward force that keeps the mass from moving in a straight line; it is the same size as centrifugal force, with the opposite sign. Centripetal force is real; centrifugal force is just an apparent force.) The mathematical expression for centrifugal acceleration (force divided by density) is

$$\text{centrifugal acceleration} = \Omega^2 r \quad (7.7)$$

where Ω is the rotation rate of Earth, equal to $2\pi/T$ where T is the length of day, and r is Earth's radius. Because the centrifugal acceleration is nearly constant in time and points outward, away from Earth's axis of rotation, we usually combine it formally with the gravitational force, which points toward Earth's center. We replace g in Eq. (7.6) with an effective gravity g , which has a weak dependence on latitude. Hereafter, we do not refer separately to the centrifugal force. The surface perpendicular to this combined force is called the *geoid*. If the ocean were not moving relative to Earth, its surface would align with the geoid.

The second term in a rotating frame of reference included in the acceleration equation (7.1) is the *Coriolis force*. When a water parcel, air parcel, bullet, hockey puck, or any other body that has little friction moves, Earth spins out from under it. By Newton's Law, the body moves in a straight line if there is no other force acting on it. As observers attached to Earth, we see the body appear to move relative to our location. In the Northern Hemisphere, the Coriolis force causes a moving body to appear to move to the right of its direction of motion (Figure S7.2b). In the Southern Hemisphere, it moves to the left.

The Coriolis force is non-zero only if the body is in motion, and is important only if the body travels for a significant period of time. Coriolis force is larger for larger velocities as well. Mathematically, the Coriolis force is

$$\text{x-momentum equation : } -2\Omega \sin\phi \, v \equiv -f \, v \quad (7.8a)$$

$$\text{y-momentum equation : } 2\Omega \sin\phi \, u \equiv f \, u \quad (7.8b)$$

$$\text{Coriolis parameter : } f \equiv 2\Omega \sin\phi \quad (7.8c)$$

where “ $\equiv$ ” denotes a definition, Ω is the rotation rate, ϕ is latitude, u is velocity in the x -direction, v is velocity in the y -direction, and where the signs are appropriate for including these terms on the left-hand side of Eq. (7.1). The *Coriolis parameter*, f , is a function of latitude and changes sign at the equator, and it has units of sec^{-1} . (The non-dimensional parameter called the *Rossby number* introduced in Section 1.2 is $Ro = 1/fT$ or $Ro = U/fL$, where U , L , and T are characteristic velocity, length, and timescales for the flow.)

7.2.4. Viscous Force or Dissipation

Fluids have viscous molecular processes that smooth out variations in velocity and slow down the overall flow. These molecular processes are very weak, so fluids can often be treated, theoretically, as “inviscid” rather than viscous. However, it is observed that turbulent fluids like the ocean and atmosphere actually act as if the effective viscosity were much larger than the molecular viscosity. *Eddy viscosity* is introduced to account for this more efficient mixing (Section 7.2.4.2).

7.2.4.1. Molecular Viscosity

We can think of molecular viscosity by considering two very different types of coexisting motion: the flow field of the fluid, and, due to their thermal energy, the random motion of molecules within the flow field. The random molecular motion carries (or advects) the larger scale velocity from one location to another, and then collisions with other molecules transfer their momentum to each other; this smooths

out the larger-scale velocity structure (Figure S7.3).

The viscous stress within a Newtonian fluid is proportional to the velocity shear. The proportionality constant is the *dynamic viscosity*, which has meter-kilogram-second (mks) units of kg/m-sec. The dynamic viscosity is the product of fluid density times a quantity called the *kinematic viscosity*, which has mks units of m²/sec. For water, the kinematic viscosity is 1.8×10^{-6} m²/sec at 0°C and 1.0×10^{-6} m²/sec at 20°C (Table S7.1). Flow is accelerated or decelerated if there is a variation in viscous stress from one location to another (Figure 7.1e).

Formally, for a *Newtonian fluid*, which is defined to be a fluid in which stress is proportional to strain (velocity shear), and if viscosity has no spatial dependence, viscous stress enters the momentum equations as

$$\begin{aligned} &\text{x-momentum dissipation} \\ &= \nu(\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 + \partial^2 u / \partial z^2) \quad (7.9) \end{aligned}$$

where ν is the molecular (kinematic) viscosity. (The dynamic viscosity is $\rho\nu$.) Molecular viscosity changes flow very slowly. Its effectiveness can be gauged by a non-dimensional parameter, the *Reynolds number*, which is the ratio of the dissipation timescale to the advective timescale: $Re = UL/\nu$. When the Reynolds number is large, the flow is nearly inviscid and most likely very turbulent; this is the case for flows governed by molecular viscosity. When Earth's rotation and hence the Coriolis term is important, the non-dimensional parameter of most interest for judging the effectiveness of dissipation is the Ekman number: $E = \nu/fH^2$. Nearly inviscid rotating flows have very small Ekman number. From matching observations and theory we know that the ocean currents dissipate energy much more quickly than we can predict using molecular viscosity. How this happens is described next.

7.2.4.2. Eddy Viscosity

Mixing at spatial scales larger than those quickly affected by molecular viscosity is generally a result of turbulence in the fluid. Turbulent motions stir the fluid, deforming and pulling it into elongated, narrow filaments. A stirred fluid mixes much faster than one that is calm and subjected only to molecular motion. We refer to the effect of this turbulent stirring/mixing on the fluid as *eddy viscosity*. For large-scale ocean circulation, the “turbulent” motions are mesoscale eddies, vertical fine structure, and so on, with spatial scales smaller than the larger scales of interest. Like molecular viscosity, eddy viscosity should be proportional to the product of turbulent speed and path length. Therefore, horizontal eddy viscosity is generally much larger than vertical eddy viscosity (Table S7.1).

To mathematically include eddy viscosity, the viscous terms in Eqs. (7.1) and (7.9) are replaced by the eddy viscosity terms:

$$\begin{aligned} &\text{x-momentum dissipation} \\ &= A_H(\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2) + A_V(\partial^2 u / \partial z^2) \quad (7.10) \end{aligned}$$

where A_H is the horizontal eddy viscosity and A_V is the vertical eddy viscosity. A_H and A_V have units of kinematic viscosity, m²/sec in mks units. (Although we often use these Cartesian coordinates, the most relevant stirring/mixing directions are along isopycnals (isentropic surfaces) and across isopycnals (*diapycnal mixing*), so the coordinate system used in Eq. (7.10) is better modeled by rotating it to have the “vertical” direction perpendicular to isopycnal surfaces, and replace A_H and A_V with eddy viscosities that are along and perpendicular to those surfaces.)

Although eddy viscosity is much larger than molecular viscosity, the ocean is nevertheless nearly inviscid, in the sense that the Reynolds number is large and the Ekman number is small even when eddy viscosities are used.

7.2.5. Mathematical Expression of Momentum Balance

The full momentum balance with spatially varying eddy viscosity and rotation is:

$$\begin{aligned} Du/Dt - fv &= \partial u/\partial t + u \partial u/\partial x \\ &+ v \partial u/\partial y + w \partial u/\partial z - fv \\ &= -(1/\rho) \partial p/\partial x + \partial/\partial x (A_H \partial u/\partial x) \\ &+ \partial/\partial y (A_H \partial u/\partial y) + \partial/\partial z (A_V \partial u/\partial z) \end{aligned} \quad (7.11a)$$

$$\begin{aligned} Dv/Dt + fu &= \partial v/\partial t + u \partial v/\partial x + v \partial v/\partial y \\ &+ w \partial v/\partial z + fu = -(1/\rho) \partial p/\partial y \\ &+ \partial/\partial x (A_H \partial v/\partial x) + \partial/\partial y (A_H \partial v/\partial y) \\ &+ \partial/\partial z (A_V \partial v/\partial z) \end{aligned} \quad (7.11b)$$

$$\begin{aligned} Dw/Dt &= \partial w/\partial t + u \partial w/\partial x + v \partial w/\partial y \\ &+ w \partial w/\partial z = -(1/\rho) \partial p/\partial z - g \\ &+ \partial/\partial x (A_H \partial w/\partial x) + \partial/\partial y (A_H \partial w/\partial y) \\ &+ \partial/\partial z (A_V \partial w/\partial z) \end{aligned} \quad (7.11c)$$

Here the standard notation “D/Dt” is the substantial derivative defined in Eq. (7.4).

The full set of equations describing the physical state of the ocean must also include the *continuity* (or *mass conservation*) equation (Section 5.1):

$$D\rho/Dt + \rho(\partial u/\partial x + \partial v/\partial y + \partial w/\partial z) = 0 \quad (7.11d)$$

If density changes are small, Eq. 7.11d is approximated as

$$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0 \quad (7.11e)$$

which is known as the *continuity* equation.

The set is completed by the equations governing changes in temperature, salinity, and density, which are presented in the following section.

7.3. TEMPERATURE, SALINITY, AND DENSITY EVOLUTION

Evolution equations for temperature and salinity — the equation of state that relates density to salinity, temperature, and pressure, and thus an evolution equation for density — complete the set of equations (7.11a–d) that describe fluid flow in the ocean.

7.3.1. Temperature, Salinity, and Density Equations

Temperature is changed by heating, cooling, and diffusion. Salinity is changed by addition or removal of freshwater, which alters the dilution of the salts. Density is then computed from temperature and salinity using the equation of state of seawater. The “word” equations for temperature, salinity, and density forcing include:

$$\begin{aligned} &\text{temperature change} \\ &+ \text{temperature advection/convection} \\ &= \text{heating/cooling term} + \text{diffusion} \end{aligned} \quad (7.12a)$$

$$\begin{aligned} &\text{salinity change} + \text{salinity advection/convection} \\ &= \text{evaporation/precipitation/runoff} \\ &\quad / \text{brine rejection} + \text{diffusion} \end{aligned} \quad (7.12b)$$

$$\begin{aligned} &\text{equation of state (dependence of density on} \\ &\quad \text{salinity, temperature, and pressure)} \end{aligned} \quad (7.12c)$$

$$\begin{aligned} &\text{density change} + \text{density advection/convection} \\ &= \text{density sources} + \text{diffusion} \end{aligned} \quad (7.12d)$$

Written in full, these are

$$\begin{aligned} DT/Dt &= \partial T/\partial t + u \partial T/\partial x + v \partial T/\partial y \\ &+ w \partial T/\partial z = Q_H/\rho c_p + \partial/\partial x (\kappa_H \partial T/\partial x) \\ &+ \partial/\partial y (\kappa_H \partial T/\partial y) + \partial/\partial z (\kappa_V \partial T/\partial z) \end{aligned} \quad (7.13a)$$

$$\begin{aligned}
 DS/Dt &= \partial S/\partial t + u \partial S/\partial x + v \partial S/\partial y + w \partial S/\partial z \\
 &= Q_S + \partial/\partial x(\kappa_H \partial S/\partial x) \\
 &\quad + \partial/\partial y(\kappa_H \partial S/\partial y) + \partial/\partial z(\kappa_V \partial S/\partial z)
 \end{aligned}
 \tag{7.13b}$$

$$\rho = \rho(S, T, p) \tag{7.13c}$$

$$\begin{aligned}
 D\rho/Dt &= \partial \rho/\partial t + u \partial \rho/\partial x + v \partial \rho/\partial y \\
 &\quad + w \partial \rho/\partial z = (\partial \rho/\partial S) DS/Dt \\
 &\quad + (\partial \rho/\partial T) DT/Dt + (\partial \rho/\partial p) Dp/Dt
 \end{aligned}
 \tag{7.13d}$$

where Q_H is the heat source (positive for heating, negative for cooling, applied mainly near the sea surface), c_p is the specific heat of seawater, and Q_S is the salinity “source” (positive for evaporation and brine rejection, negative for precipitation and runoff, applied at or near the sea surface). κ_H and κ_V are the horizontal and vertical eddy diffusivities, analogous to the horizontal and vertical eddy viscosities in the momentum equations (7.11a–d) (Table S7.1 located on the textbook Web site). The full equation of state appears in Eq. (7.13c), from which the evolution of density in terms of temperature and salinity change can be computed (Eq. 7.13d). The coefficients for the three terms in Eq. (7.13d) are the haline contraction coefficient, the thermal expansion coefficient, and the

adiabatic compressibility, which is proportional to the inverse of sound speed (Chapter 3).

7.3.2. Molecular and Eddy Diffusivity

The molecular diffusivity κ for each substance depends on the substance and the fluid. The molecular diffusivity of salt in seawater is much smaller than that for heat (Table S7.1). This difference results in a process called “double diffusion” (Section 7.4.3). *Eddy diffusivity* is the equivalent of eddy viscosity for properties like heat and salt. A globally averaged vertical eddy diffusivity of $\kappa_V = 1 \times 10^{-4} \text{ m}^2/\text{sec}$ accounts for the observed average vertical density structure (Section 7.10.2; Munk, 1966). However, the directly observed vertical (or diapycnal) eddy diffusivity in most of the ocean is a factor of 10 lower: $\kappa_V \sim 1 \times 10^{-5} \text{ m}^2/\text{sec}$, implying that there are regions of much higher diffusivity to reach the global average. Measurements show huge enhancements of diapycnal eddy diffusivity in bottom boundary regions, especially where topography is rough (Figure 7.2) (Polzin, Toole, Ledwell, & Schmitt, 1997; Kunze et al., 2006), and on continental shelves where tidal energy is focused (Lien & Gregg, 2001). In the surface layer, eddy diffusivities and eddy viscosities are also much greater than the Munk value (e.g., Large, McWilliams, & Doney, 1994) (Section S7.4.1). Horizontal eddy diffusivities κ_H are estimated to be between 10^3 and

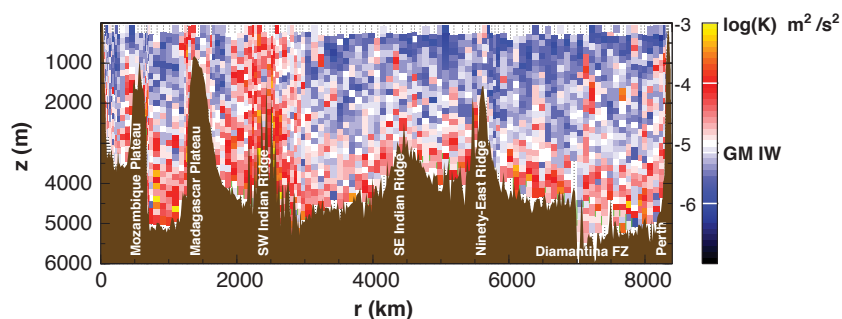


FIGURE 7.2 Observed diapycnal diffusivity (m^2/s^2) along 32°S in the Indian Ocean, which is representative of other ocean transects of diffusivity. See Figure S7.4 for diffusivity profiles. This figure can also be found in the color insert. ©American Meteorological Society. Reprinted with permission. Source: From Kunze et al. (2006).

$10^4 \text{ m}^2/\text{sec}$, with large spatial variability (e.g., Figure 14.17). κ_H is much larger than κ_V .

7.4. MIXING LAYERS

Mixing occurs throughout the ocean. While weak, mixing is essential for maintaining the observed stratification and can regulate the strength of some parts of the circulation.

7.4.1. Surface Mixed Layer

The surface layer (Section 4.2.2) is forced directly by the atmosphere through surface wind stress and buoyancy (heat and freshwater) exchange. For a surface layer that is initially stably stratified (Figure 7.3a), sufficiently large wind stress will create turbulence that mixes and creates a substantially uniform density or mixed layer (Figure 7.3b). This typically results in a discontinuity in properties at the mixed layer base.

The upper layer can also be mixed by buoyancy loss through the sea surface, increasing the density of the top of the surface layer and causing it to overturn (*convect*) to a greater depth (Figure 7.3c–e).

This type of mixed layer typically has no discontinuity in density at its base. Heat or freshwater gain decreases the density of the top of the surface layer, resulting in a more stably stratified profile. If the wind then mixes it, the final mixed layer is shallower than the initial mixed layer (Figure 7.3f–h). (Mixed layer observations typically show much more vertical structure than might be expected from these simple ideas.)

The thickest mixed layers occur at the end of winter (Figure 4.5), after an accumulation of months of cooling that deepens the mixed layer and increases its density. For large-scale oceanographic studies, these end-of-winter mixed layers set the properties that are subducted into the ocean interior (Section 7.8.5).

7.4.2. Bottom Mixed Layers

Near the ocean bottom turbulence, and hence mixing, can be generated by currents or current shear caused by the interaction with the bottom. In shallow (e.g., coastal) waters, complete mixing of the water column occurs if the depth is shallow enough and the tidal currents are fast enough (see reviews in Simpson, 1998 and Brink, 2005 and more extended discussion in Section S7.4.2 on the textbook Web site). At longer timescales on the shelf, a bottom Ekman layer can develop in which frictional and Coriolis forces balance (Ekman, 1905 and Section 7.5.3), with the bottom slope also affecting the layer.

Enhanced turbulence in a bottom boundary layer can be created by movement of water across rough topography and by breaking of internal waves that reflect off the topography and result in higher eddy diffusivity values (Figure 7.2). This can create “steppy” vertical profiles near the bottom some distance from the mixing site (Figure S7.6a located on the textbook Web site).

Bottom currents due to density differences can also cause mixing. One example is a turbidity current down an underlying bottom slope (Section 2.6). Another example is the overflow of dense water across a sill, as seen at the Strait of Gibraltar (Chapter 9). The dense water flows down the continental slope as a *plume*, mixing vigorously with the lighter water around it (Figure S7.6b). This turbulent process is called *entrainment*. Density differences due to the injection of lighter water into the ocean also cause mixing and entrainment. An example is hot hydrothermal water injected at mid-ocean ridges and hotspots that entrain ambient waters as the plumes rise.

7.4.3. Internal Mixing Layers

In the interior of the ocean (i.e., away from boundaries), continuous profiling instruments have shown that vertical profiles of water properties — temperature and salinity, and hence density — are often not smooth (Figure 7.3i)

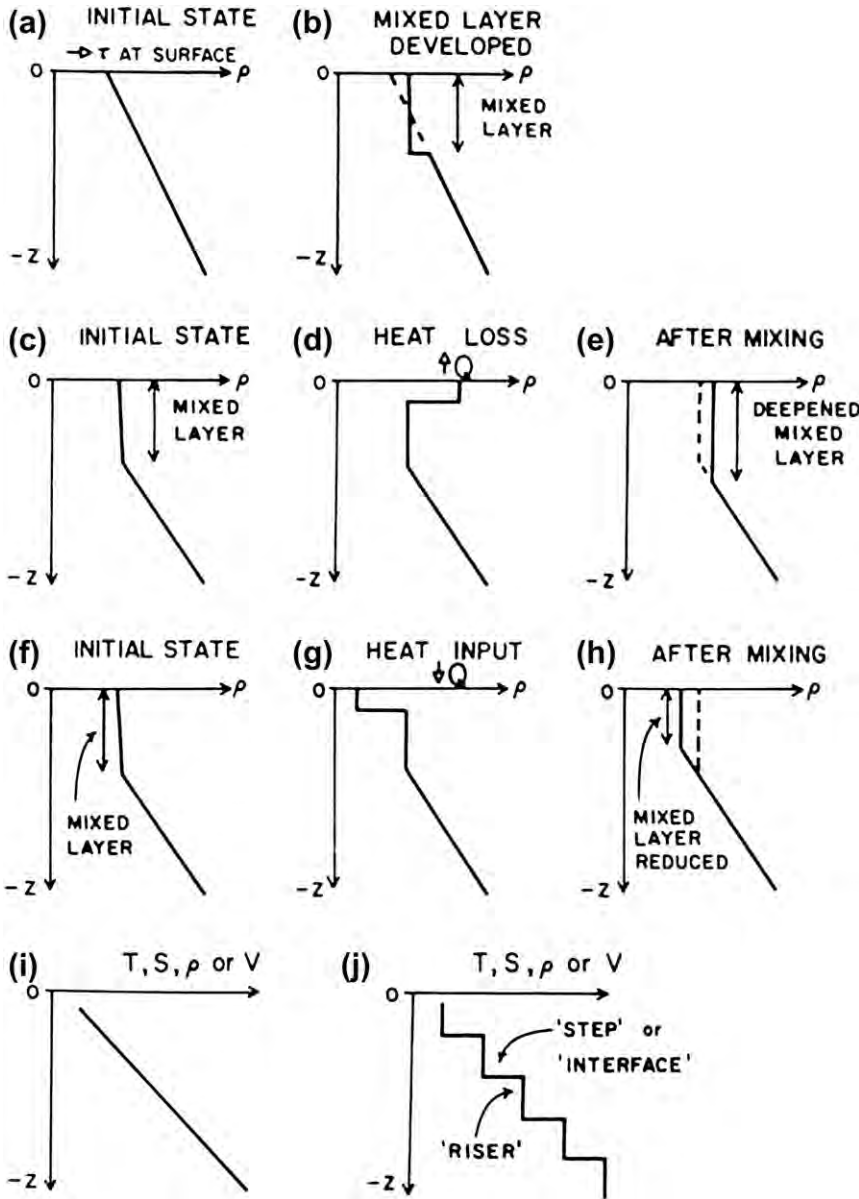


FIGURE 7.3 Mixed layer development. (a, b) An initially stratified layer mixed by turbulence created by wind stress; (c, d, e) an initial mixed layer subjected to heat loss at the surface which deepens the mixed layer; (f, g, h) an initial mixed layer subjected to heat gain and then to turbulent mixing presumably by the wind, resulting in a thinner mixed layer; (i, j) an initially stratified profile subjected to internal mixing, which creates a stepped profile. Notation: τ is wind stress, Q is heat (buoyancy).

but “stepped” (Figure 7.3j). Turbulence and/or double diffusion mix the water column internally and can create such steps.

Breaking internal waves (Chapter 8) can create internal mixing (Section 8.4; Rudnick et al., 2003). Vertical shear from other sources can also result

in turbulence. On the other hand, vertical stratification stabilizes the mixing. One way to express this trade-off is through a non-dimensional parameter called the *Richardson number* (Ri):

$$Ri = N^2 / (\partial u / \partial z)^2 \quad (7.14)$$

$$N^2 = -g (\partial \rho / \partial z) / \rho_0 \quad (7.15)$$

where N is the Brunt-Väisälä frequency (Section 3.5.6) and the vertical shear of the horizontal speed is $(\partial u / \partial z)$. If the Richardson number is small, the stratification is weak and the shear is large, so we expect mixing to be vigorous.

Stirring between two horizontally adjacent waters with strongly contrasting temperature and salinity results in *interleaving* or *fine structure*, with layering of one to tens of meters on both sides of the front. A much smaller scale of vertical structure, on the order of centimeters (*microstructure*), is associated with the actual mixing at the interfaces between the interleaving layers.

Heat diffuses about 100 times faster than salt (Table S7.1). *Double diffusion* arises from these differing molecular diffusivities, acting at scales of centimeters to meters, and can also create well-mixed internal layers. When warm, salty water lies above cold, fresh water, the saltier water becomes denser and tends to sink into the lower layer and vice versa (Figure S7.7a). The alternating columns are called *salt fingers*. Lateral diffusion occurs between the “fingers” and produces a uniform layer that may be meters to tens of meters thick in the ocean. When cold, fresh water lies above warm, salty water (Figure S7.7b), the fresher upper layer becomes warmer and rises within the upper layer; this is called the *diffusive* form of double diffusion. Salt fingering effects are observed in the ocean where there are strong contrasts in salinity, for instance, where salty Mediterranean Water enters the Atlantic (Figure S7.7c). Diffusive interfaces are observed in high latitude regions with dichothermal layers (Sections 4.2, 4.3.2 and Figure S7.7d).

7.5. RESPONSE TO WIND FORCING

The wind blows over the sea surface exerting stress and causing the water to move within the

top 50 m. Initially the wind excites small capillary waves that propagate in the direction of the wind. Continued wind-driven momentum exchange excites a range of surface waves (Chapter 8). The net effect of this input of atmospheric momentum is a stress on the ocean (*wind stress*). For timescales of about a day and longer, Earth’s rotation becomes important and the Coriolis effect enters in, as described in the following subsections.

7.5.1. Inertial Currents

At timescales of a day or so (after build-up of surface waves and possibly Langmuir circulation), the ocean responds to a wind stress impulse with transient motions known as “inertial currents.” These are a balance of the Coriolis force and the time derivatives of the initial horizontal velocities caused by the wind stress. In the Northern Hemisphere, the water particles trace out clockwise circles (Figure S7.8a). In the Southern Hemisphere, inertial currents are counterclockwise.

(Mathematically, inertial currents are the solution of

$$\partial u / \partial t = fv \quad (7.16a)$$

$$\partial v / \partial t = -fu \quad (7.16b)$$

which is taken from Eq. (7.11a and b) assuming that advection, pressure gradient forces, and dissipation are very small.)

Inertial currents are often observed in surface drifter trajectories and surface velocity moorings in the wake of a storm (Figure 7.4). Inertial periods are often very close to tidal periods, so separating tidal and inertial effects in time series is sometimes difficult.

After the wind starts to blow impulsively, the current will initially oscillate around and then, after several days, settle frictionally to a steady flow at an angle to the wind (Figure S7.8b from Ekman, 1905). This becomes the surface Ekman velocity (Section 7.5.3).

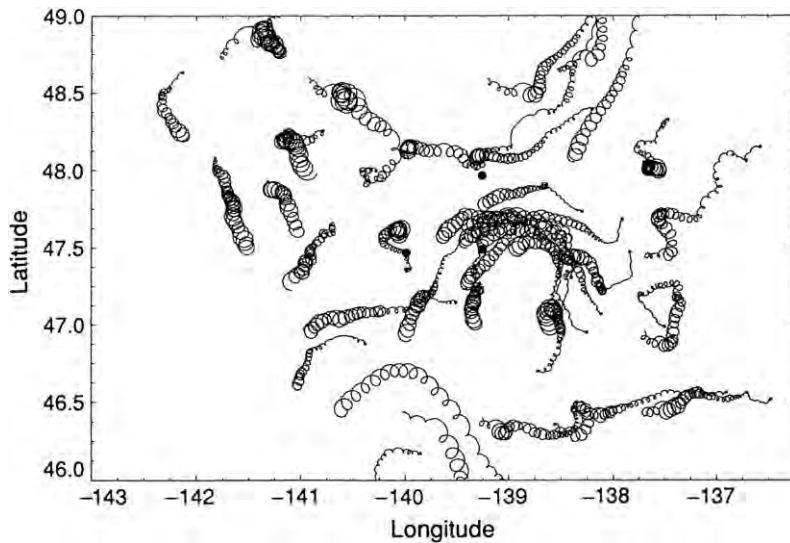


FIGURE 7.4 Observations of near-inertial currents. Surface drifter tracks during and after a storm. ©American Meteorological Society. Reprinted with permission. Source: From d'Asaro et al. (1995). See Figure S7.8 for schematics of inertial currents and Ekman's (1905) original hodograph.

7.5.2. Langmuir Circulation

"Langmuir circulation" (LC) is another transient response to impulsive wind forcing (Langmuir, 1938). A lengthier discussion with illustrations is provided in Section S7.5.2 in the online supplementary material. LCs are visually evident as numerous long parallel lines or streaks of flotsam ("windrows") that are mostly aligned with the wind (Figure S7.9). The streaks are formed by the convergence caused by helical vortices with a typical depth and horizontal spacing of 4–6 m and 10–50 m, but they can range up to several hundred meters horizontal separation and up to two to three times the mixed layer depth (Figure S7.10). Alternate cells rotate in opposite directions, causing convergence and divergence between alternate pairs of cells. The cells can be many kilometers long. Langmuir circulations generally occur only for wind speeds greater than 3 m/sec and appear within a few tens of minutes of wind onset. The mechanism for producing Langmuir circulation is still a matter of study and beyond the scope of this text. See Smith (2001) and Thorpe (2004) for further discussions.

7.5.3. Ekman Layers

Wind stress is communicated to the ocean surface layer through viscous (frictional) processes that extend several tens of meters into the ocean. For timescales longer than a day, the response is strongly affected by Coriolis acceleration. This wind-driven frictional layer is called the *Ekman layer*. The physical processes in an Ekman layer include only friction (eddy viscosity) and Coriolis acceleration. Velocity in the Ekman layer is strongest at the sea surface and decays exponentially downward, disappearing at a depth of about 50 m. It coexists with, but is not the same as, the mixed layer depth or euphotic zone depth.

The two most unusual characteristics of an Ekman layer (compared with a frictional flow that is not rotating) are (1) the horizontal velocity vector spirals with increasing depth (Figure 7.5) and (2) the net transport integrated through the Ekman layer is *exactly* to the right of the wind in the Northern Hemisphere (left in the Southern Hemisphere).

The surface water in an Ekman layer moves at an angle to the wind because of Coriolis acceleration. If eddy viscosity is independent

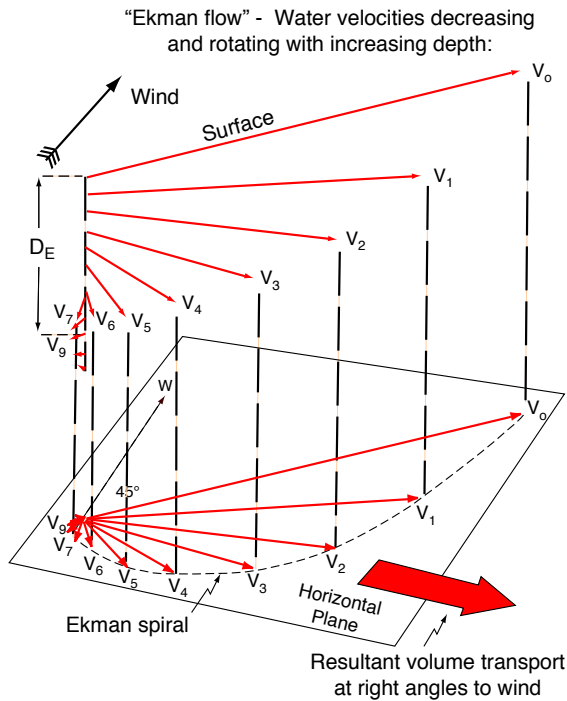


FIGURE 7.5 Ekman layer velocities (Northern Hemisphere). Water velocity as a function of depth (upper projection) and Ekman spiral (lower projection). The large open arrow shows the direction of the total Ekman transport, which is perpendicular to the wind.

of depth, the angle is 45 degrees to the right of the wind in the Northern Hemisphere; otherwise it differs somewhat from 45 degrees. Due to the frictional stress proportional to the eddy viscosity A_V , each layer from the surface on down accelerates the next layer below to the right (Northern Hemisphere) and has a weaker velocity than the layer above it. The complete structure is a decaying “spiral.” If the velocity arrows are projected onto a horizontal plane, their tips form the Ekman spiral (Figure 7.5).

The Ekman layer depth is the e-folding depth of the decaying velocity:

$$D_E = (2A_V/f)^{1/2} \quad (7.17)$$

Using a constant eddy viscosity of $0.05 \text{ m}^2/\text{sec}$ from within the observed range (Section 7.5.5), the Ekman layer depths at latitudes 10, 45, and 80 degrees are 63, 31, and 26 m, respectively. The vertically integrated horizontal velocity in the Ekman layer is called the *Ekman transport*:

$$U_E = \int u_E(z) dz \quad (7.18a)$$

$$V_E = \int v_E(z) dz \quad (7.18b)$$

where u_E and v_E are the eastward and northward velocities in the Ekman layer, and U_E and V_E are the associated Ekman transports. (Ekman “transport” has units of depth times velocity, hence m^2/sec , rather than area times velocity.) Ekman transport in terms of the wind stress is derived from Eq. (7.11):

$$U_E = \tau^{(y)}/(\rho f) \quad (7.19a)$$

$$V_E = -\tau^{(x)}/(\rho f) \quad (7.19b)$$

where $\tau^{(x)}$ and $\tau^{(y)}$ are the wind stresses positive in the east and north directions, assuming no time acceleration, advection, or pressure gradient force, and setting the eddy friction stress at the sea surface equal to the wind stress. The Ekman transport is *exactly* perpendicular and to the right (left) of the wind in the Northern (Southern) Hemisphere (large arrow in Figure 7.5). For applications of Ekman layers to general circulation (Sections 7.8 and 7.9), only the Ekman transport matters. Thus, the actual eddy viscosity and Ekman layer thickness are unimportant.

Bottom Ekman layers that are 50 to 100 m thick can develop if there is a flow along the bottom. In shallow water, the top and bottom Ekman layers can overlap, so that the right-turning tendency in the top layer (Northern Hemisphere) will overlap the left-turning tendency in the bottom layer. If there is a wind stress at the top surface that would produce an Ekman layer of depth D_E in deep water, then

in water of depth h , the approximate angle α between the wind and the surface flow is as listed in Table S7.2. As water depth decreases, the net flow is more in the direction of the wind.

7.5.4. Ekman Transport Convergence and Wind Stress Curl

When the wind stress varies with position so that Ekman transport varies with position, there can be a convergence or divergence of water within the Ekman layer. Convergence results in downwelling of water out of the Ekman layer. Divergence results in upwelling into the Ekman layer. This is the mechanism that connects the frictional forcing by wind of the surface layer to the interior, geostrophic ocean circulation (Section 7.8).

The vertical velocity w_E at the base of the Ekman layer is obtained from the divergence of the Ekman transport, by vertically integrating the continuity equation Eq. (7.11e) over the depth of the Ekman layer:

$$\begin{aligned} (\partial U_E / \partial x + \partial V_E / \partial y) &= \nabla \cdot \mathbf{U}_E \\ &= -(w_{\text{surface}} - w_E) = w_E \end{aligned} \quad (7.20)$$

where $\mathbf{U}_E$ is the horizontal vector Ekman transport (Eq. 7.18) and it is assumed that the vertical velocity at the sea surface, w_{surface} , is 0. When Eq. (7.20) is negative, the transport is convergent and there must be downwelling (w_E at the base of the Ekman layer is negative). The relation of Ekman transport divergence to the wind stress from Eq. (7.19a,b) is

$$\begin{aligned} \nabla \cdot \mathbf{U}_E &= \partial / \partial x (\tau^{(y)} / (\rho f)) - \partial / \partial y (\tau^{(x)} / (\rho f)) \\ &= \mathbf{k} \cdot \nabla \times (\boldsymbol{\tau} / \rho f) \end{aligned} \quad (7.21)$$

where $\boldsymbol{\tau}$ is the vector wind stress and $\mathbf{k}$ is the unit vector in the vertical direction. Therefore, in the Northern Hemisphere ($f > 0$), upwelling into the Ekman layer results from positive wind stress curl, and downwelling results

from negative wind stress curl. Downwelling is referred to as *Ekman pumping*. Upwelling is sometimes referred to as Ekman suction.

A global map of wind stress curl was shown in Figure 5.16d, and is referred to frequently in subsequent chapters because of its importance for Ekman pumping/suction, although the mapped quantity should include the Coriolis parameter, f , to be related directly to upwelling and downwelling.

Equatorial upwelling due to Ekman transport results from the westward wind stress (trade winds). These cause northward Ekman transport north of the equator and southward Ekman transport south of the equator. This results in upwelling along the equator, even though the wind stress curl is small, more or less because of the Coriolis parameter dependence in Eq. (7.21).

The coastline is the other place where Ekman transport divergence or convergence can occur, and it is *not* included in Eq. (7.21) because this divergence is due to the boundary condition at the coast and not wind stress curl. If the wind blows along the coast, then Ekman transport is perpendicular to the coast, so there must be either downwelling or upwelling at the coast to feed the Ekman layer (Figure 7.6). This is one mechanism for creation of coastal upwelling and subtropical eastern boundary current systems (Section 7.9).

7.5.5. Observations of Ekman Response and Wind Forcing

The Ekman theory has major consequences for wind-driven ocean circulation. Thus it is important to confirm and refine Ekman's theory with ocean observations. Observations of Ekman response are difficult because of the time dependence of the wind. California Current observations produced an easily visible Ekman-like response because the wind direction was relatively steady (Figure 7.7 and Chereskin, 1995).

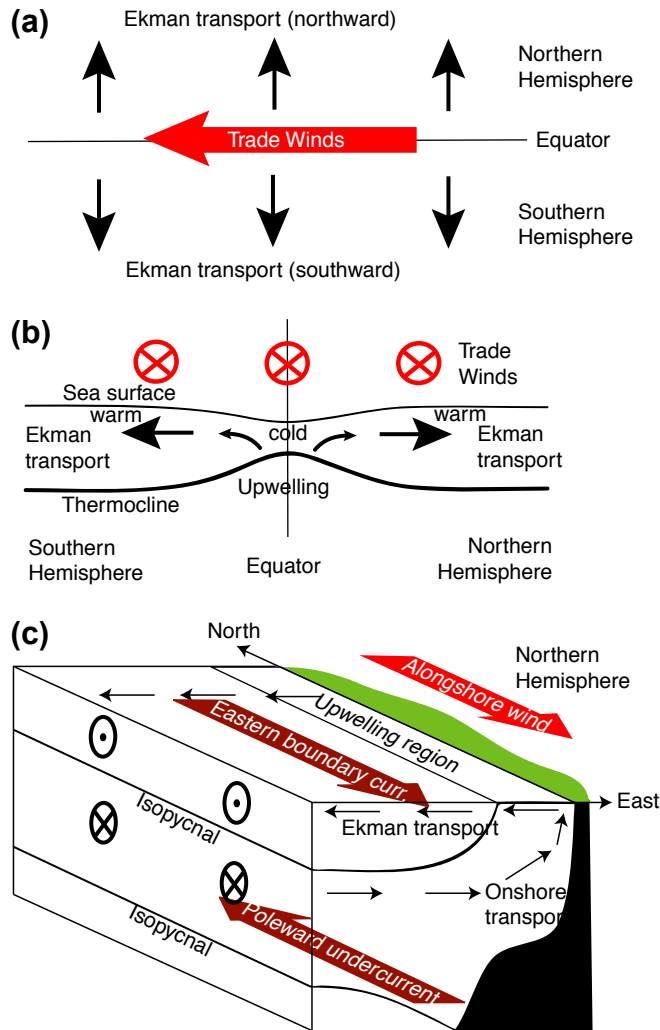


FIGURE 7.6 (a) Ekman transport divergence near the equator driven by easterly Trade Winds. (b) The effect of equatorial Ekman transport divergence on the surface height, thermocline, and surface temperature. (c) Coastal upwelling system due to an alongshore wind with offshore Ekman transport (Northern Hemisphere).

An Ekman response to the wind for a large part of the Pacific Ocean is apparent in the average 15 m velocity from surface drifters (Figure 7.8). Velocities are to the right of the wind stress in the Northern Hemisphere and to the left in the Southern Hemisphere.

7.6. GEOSTROPHIC BALANCE

7.6.1. Pressure Gradient Force and Coriolis Force Balance

Throughout most of the ocean at timescales longer than several days and at spatial scales

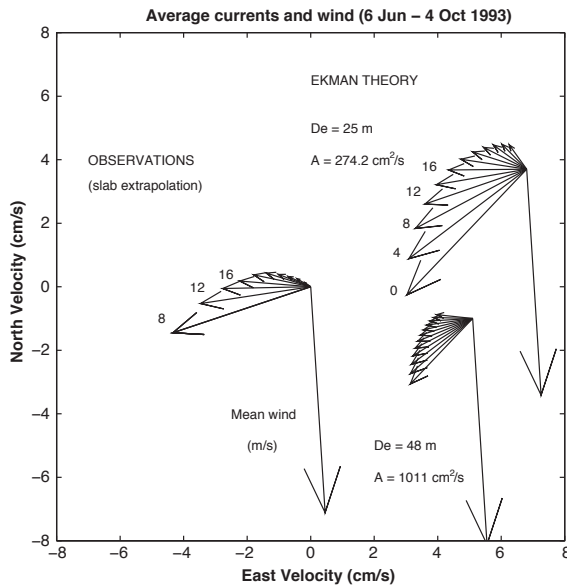


FIGURE 7.7 Observations of an Ekman-like response in the California Current region. Observed mean velocities (left) and two theoretical Ekman spirals (offset) using different eddy diffusivities (274 and $1011 \text{ cm}^2/\text{s}$). The numbers on the arrows are depths. The large arrow is the mean wind. See Figure S7.14 for the progressive vector diagram. Source: From Chereskin (1995).

longer than several kilometers, the balance of forces in the horizontal is between the pressure gradient and the Coriolis force. This is called “geostrophic balance” or *geostrophy*.

In a “word” equation, geostrophic balance is horizontal Coriolis acceleration

$$= \text{horizontal pressure gradient force} \quad (7.22)$$

This is illustrated in Figure 7.9. The pressure gradient force vector points from high pressure to low pressure. In a non-rotating flow, the water would then move from high to low pressure. However, with rotation, the Coriolis force exactly opposes the pressure gradient force, so that the net force is zero. Thus, the water parcel does not accelerate (relative to Earth). The parcel moves exactly perpendicular to both the pressure gradient force and the Coriolis force. A heuristic way to remember the direction of geostrophic flow is to think of the pressure gradient force pushing the water parcel from high to low pressure, but Coriolis force moves the parcel off to the right (Northern Hemisphere) or the left (Southern Hemisphere).

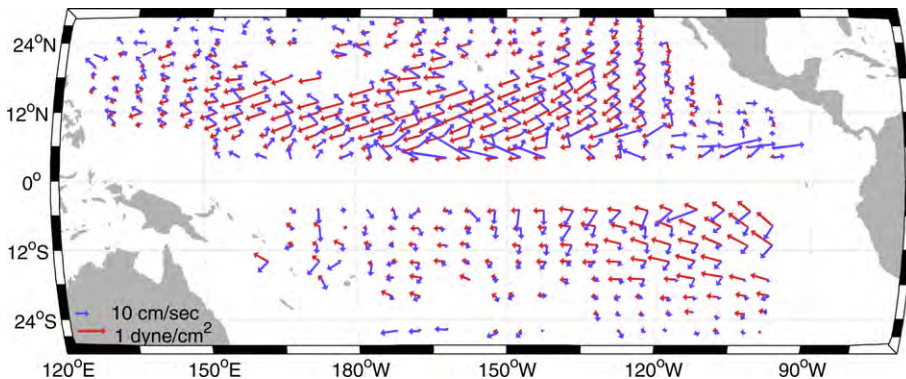


FIGURE 7.8 Ekman response. Average wind vectors (blue) and average ageostrophic current at 15 m depth (red). The current is calculated from 7 years of surface drifters drogued at 15 m, with the geostrophic current based on average density data from Levitus et al. (1994a) removed. (No arrows were plotted within 5 degrees of the equator because the Coriolis force is small there.) This figure can also be found in the color insert. ©American Meteorological Society. Reprinted with permission. Source: From Ralph and Niiler (1999).

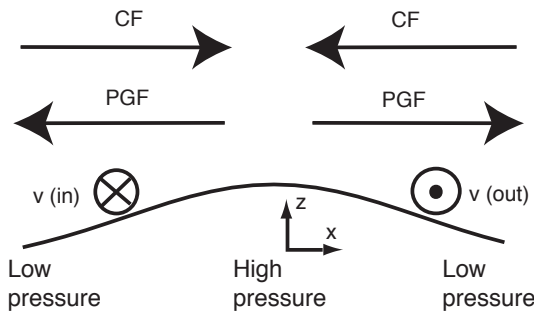


FIGURE 7.9 Geostrophic balance: horizontal forces and velocity. PGF = pressure gradient force. CF = Coriolis force. v = velocity (into and out of page). See also Figure S7.17.

The vertical force balance that goes with geostrophy is *hydrostatic balance* (Section 3.2). The vertical pressure gradient force, which points upward from high pressure to low pressure, is balanced by gravity, which points downward.

The mathematical expression of geostrophy and hydrostatic balance, from Eq. (7.11), is

$$-fv = -(1/\rho)\partial p/\partial x \quad (7.23a)$$

$$fu = -(1/\rho)\partial p/\partial y \quad (7.23b)$$

$$0 = -\partial p/\partial z - \rho g \quad (7.23c)$$

An alternate form for Eq. (7.23c), used for dynamic height calculations (Section 7.6.3), is

$$0 = -\alpha \partial p/\partial z - g \quad (7.23d)$$

where α is specific volume. From Eq. (7.23a and b), if the Coriolis parameter is approximately constant ($f = f_0$), the geostrophic velocities are approximately non-divergent:

$$\partial u/\partial x + \partial v/\partial y = 0 \quad (7.23e)$$

Formally in fluid dynamics, such a non-divergent velocity field can be written in terms of a *streamfunction* ψ :

$$u = -\partial\psi/\partial y \text{ and } v = \partial\psi/\partial x \quad (7.23f)$$

From Eqs. (7.23a and b) the streamfunction for geostrophic flow is $\psi = p/(f_0\rho_0)$. Therefore, maps of pressure distribution (or its proxies like dynamic height, steric height, or geopotential anomaly; Section 7.6.2), are maps of the geostrophic streamfunction, and flow approximately follows the mapped contours.

Geostrophic balance is intuitively familiar to those with a general interest in weather reports. Weather maps show high and low pressure regions around which the winds blow. Low pressure regions in the atmosphere are called *cyclones*. Flow around low pressure regions is thus called *cyclonic* (counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere). Flow around high-pressure regions is called *anticyclonic*.

In the ocean, higher pressure is caused by a higher mass of water lying above the observation depth. At the “sea surface,” pressure differences are due to an actual mounding of water relative to the geoid. Over the complete width of the Atlantic or Pacific Ocean anticyclonic gyres, the total contrast in sea surface height is about 1 m.

The geostrophic velocities at the sea surface could be calculated if the appropriately time-averaged sea surface height were known (as yet not possible for the time mean, but definitely possible from satellite altimetry for variations from the mean). The geostrophic velocity at the sea surface in terms of sea surface height η above a level surface is derived from Eqs. (7.23a and b):

$$-fv = -g\partial\eta/\partial x \quad (7.24a)$$

$$fu = -g\partial\eta/\partial y \quad (7.24b)$$

To calculate the horizontal pressure difference below the sea surface, we have to consider both the total height of the pile of water above our observation depth and also its density, since the total mass determines the actual pressure at our observation depth (Figure 7.10 in this

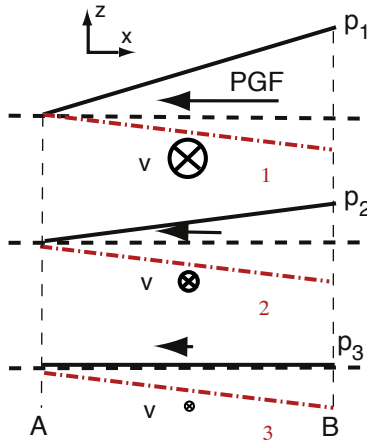


FIGURE 7.10 Geostrophic flow and thermal wind balance: schematic of change in pressure gradient force (PGF) with depth. The horizontal geostrophic velocity v is into the page for this direction of PGF and is strongest at the top, weakening with depth, as indicated by the circle sizes. Density (dash-dot) increases with depth, and isopycnals are tilted. With the sea surface at B higher than at A, the PGF at the sea surface (h_1) is to the left. The PGF decreases with increasing depth, as indicated by the flattening of the isobars p_2 and p_3 .

chapter and Figure S7.19 on the textbook Web site). The variation in geostrophic flow with depth (the *geostrophic velocity shear*) is therefore proportional to the difference in density of the two water columns on either side of our observation location. The relation between the geostrophic velocity shear and the horizontal change (gradient) in density is called the *thermal wind relation*.

The thermal wind relation is illustrated in Figure 7.10. The sea surface is sloped, with surface pressure higher to the right. This creates a pressure gradient force to the left, which drives a surface geostrophic current into the page (Northern Hemisphere). The density ρ increases with depth, and the isopycnals are tilted. Therefore the geostrophic velocity changes with depth because the pressure gradient force changes with depth due to the tilted isopycnals.

A useful rule of thumb for geostrophic flows that are surface-intensified is that, when facing downstream in the Northern Hemisphere, the “light/warm” water is to your right. (In the Southern Hemisphere, the light water is to the left when facing downstream.) It can be useful to memorize the example for the Gulf Stream recalling that the current flows eastward with warm water to the south. Geostrophic flow with vertical shear, which requires sloping isopycnals, is often called baroclinic. Geostrophic flow without any vertical shear is often called barotropic. Barotropic flow is driven only by horizontal variations in sea surface height. Most oceanic geostrophic flows have both barotropic and baroclinic components.

Mathematically, the thermal wind relations are derived from the geostrophic and hydrostatic balance Eq. (7.23):

$$-f\partial v/\partial z = (g/\rho_0)\partial\rho/\partial x \quad (7.25a)$$

$$f\partial u/\partial z = (g/\rho_0)\partial\rho/\partial y \quad (7.25b)$$

(Here we have used the *Boussinesq approximation*, where ρ can be replaced by the constant ρ_0 in the x and y momentum equations, whereas the fully variable density ρ must be used in the hydrostatic balance equation.)

To calculate geostrophic velocity, we must know the absolute horizontal pressure difference between two locations. If we have only the density distribution, we can calculate only the geostrophic shear. To convert these relative currents into absolute currents, we must determine or estimate the absolute current or pressure gradient at some level (*reference level*). A common, but usually inaccurate, referencing approach has been to assume (without measuring) that the absolute current is zero at some depth (*level of no motion*). In the next subsection, we introduce the “dynamic” method widely used to calculate geostrophic velocities (shear), and continue the discussion of reference velocity choices.

7.6.2. Geopotential and Dynamic Height Anomalies and Reference Level Velocities

Historically and continuing to the present, it has been too difficult and too expensive to instrument the ocean to directly observe velocity everywhere at every depth. Density profiles, which are much more widely and cheaply collected, are an excellent data set for estimating geostrophic velocities using the thermal wind relations and estimates of a reference level velocity. To calculate geostrophic velocity shear from density profiles, oceanographers have created two closely related functions, *geopotential anomaly* and *dynamic height*, whose horizontal gradients represent the horizontal pressure gradient force. Another closely related concept, *steric height*, is used to study variations in sea level. The gradient of the *geopotential*, Φ , is in the direction of the local force due to gravity (modified to include centrifugal force). Geopotential is defined from hydrostatic balance (Eq. 7.23c) as

$$d\Phi = g dz = -\alpha dp \quad (7.26a)$$

where α is specific volume. The units of geopotential are m^2/sec^2 or J/kg . For two isobaric surfaces p_2 (upper) and p_1 (lower), the geopotential is

$$\Phi = g \int dz = g(z_2 - z_1) = -\int \alpha dp \quad (7.26b)$$

Geopotential height is defined as

$$\begin{aligned} Z &= (9.8 \text{ m s}^{-2})^{-1} \int g dz \\ &= -(9.8 \text{ m s}^{-2})^{-1} \int \alpha dp \end{aligned} \quad (7.26c)$$

and is nearly equal to geometric height. This equation is in mks units; if centimeter-gram-second (cgs) units are used instead, the multiplicative constant would change from 9.8 m s^{-2} to 980 cm s^{-2} . Most practical calculations,

including common seawater computer subroutines, use the specific volume anomaly

$$\delta = \alpha(S, T, p) - \alpha(35, 0, p) \quad (7.26d)$$

to compute the geopotential anomaly

$$\Delta\Phi = -\int \delta dp. \quad (7.26e)$$

The geopotential height anomaly is then defined as

$$Z' = -(9.8 \text{ m s}^{-2})^{-1} \int \delta dp. \quad (7.26f)$$

Geopotential height anomaly is effectively identical to *steric height anomaly*, which is defined by Gill and Niiler (1973) as

$$h' = -(1/\rho_0) \int \rho dz \quad (7.27a)$$

in which the *density anomaly* $\rho' = \rho - \rho_0$. Using hydrostatic balance and defining ρ_0 as $\rho(35, 0, p)$, Eq. (7.27a) is equivalent to Tomczak and Godfrey's (1994) steric height (anomaly)

$$h' = \int \delta \rho_0 dz \quad (7.27b)$$

which can be further manipulated to yield

$$h' = (1/g) \int \delta dp. \quad (7.27c)$$

This is nearly identical to the geopotential height anomaly in Eq. (7.26f), differing only in the appearance of a standard quantity for g . In SI units, steric height is in meters.

Dynamic height, D , is closely related to geopotential, Φ , differing only in sign and units of reporting. Many modern publications and common computer subroutines do not distinguish between dynamic height and geopotential anomaly. The unit traditionally used for dynamic height is the dynamic meter:

$$1 \text{ dyn m} = 10 \text{ m}^2/\text{sec}^2. \quad (7.28a)$$

Therefore dynamic height reported in dynamic meters is related to geopotential anomaly as

$$\Delta D = -\Delta\Phi/10 = \int \delta \, dp/10. \quad (7.28b)$$

Its relation to the geopotential height and steric height anomalies is

$$10 \, \Delta D = -9.8 \, Z' = gh'. \quad (7.28c)$$

The quantities ΔD and Z' are often used interchangeably, differing only by 2%. With use of the dynamic meter, maps of dynamic topography are close to the actual geometric height of an isobaric surface relative to a level surface; a horizontal variation of, say, 1 dyn m, means that the isobaric surface has a horizontal depth variation of about 1 m. Note that the geopotential height anomaly more closely reflects the actual height variation, so a variation of 1 dyn m would be an actual height variation closer to 1.02 m.

Geostrophic velocities at one depth relative to those at another depth are calculated using Eq. (7.25) with geopotential anomalies, steric height anomalies, or dynamic heights. In SI units, and using dynamic meters for dynamic height, the difference between the northward velocity v and eastward velocity u at the pressure surface p_2 relative to the pressure surface p_1 is

$$\begin{aligned} f(v_2 - v_1) &= 10 \, \partial \Delta D / \partial x = -\partial \Delta\Phi / \partial x \\ &= g\delta h' / \partial x \end{aligned} \quad (7.29a)$$

$$\begin{aligned} f(u_2 - u_1) &= -10 \, \partial \Delta D / \partial y = \partial \Delta\Phi / \partial y \\ &= -g\delta h' / \partial y \end{aligned} \quad (7.29b)$$

where the dynamic height or geopotential anomalies are integrated vertically from p_1 to p_2 . The surface p_1 is the *reference level*. (Comparison of Eq. 7.29 with Eq. 7.23 shows that the dynamic height and geopotential anomalies are streamfunctions for the difference between geostrophic flows from one depth to another.)

How is the velocity at the reference level chosen? Since the strength of ocean currents

decreases from the surface downward in many (but not all) regions, for practical reasons, a deep level of no motion has often been presumed. A much better alternative is to use a “level of known motion” based on direct velocity observations. Satellite altimetry by itself is insufficient for the ocean’s mean flow field because the spatial variations of Earth’s geoid are vastly larger than the ocean’s sea-surface height variations; the **GR**avity and Earth Climate Experiment (GRACE) is helping to resolve this geoid problem. Modern practice requires that the flow field that is defined by many density profiles must satisfy overall constraints such as mass conservation. The constraints then help narrow the choices of reference level velocities, which can be done formally (see Wunsch, 1996). Ocean state estimation (data assimilation), which merges observations with an ocean model, is currently the focus of most activity for construction of velocity fields from density profiles.

As an example of the geostrophic method, we calculate dynamic height and velocity profiles from density profiles across the Gulf Stream (Figure 7.11 in this chapter and Table S7.3 on the textbook Web site). The isopycnals sloping upward toward the north between 38° and 39°N mark the Gulf Stream (Figure 7.11a). The geostrophic velocity profile is calculated between stations “A” and “B” relative to an arbitrary level of no motion at 3000 m. Station A has lower specific volume (higher potential density) than station B (Figure 7.11b). The surface dynamic height at A is therefore lower than at B and the surface pressure gradient force is toward the north, from B to A. Therefore, the geostrophic velocity at the midpoint between the stations (Figure 7.11d) is eastward and is largest at the sea surface. This means that the sea surface must tilt downward from B to A. The vertical shear is largest in the upper 800 m where the difference in dynamic heights is largest.

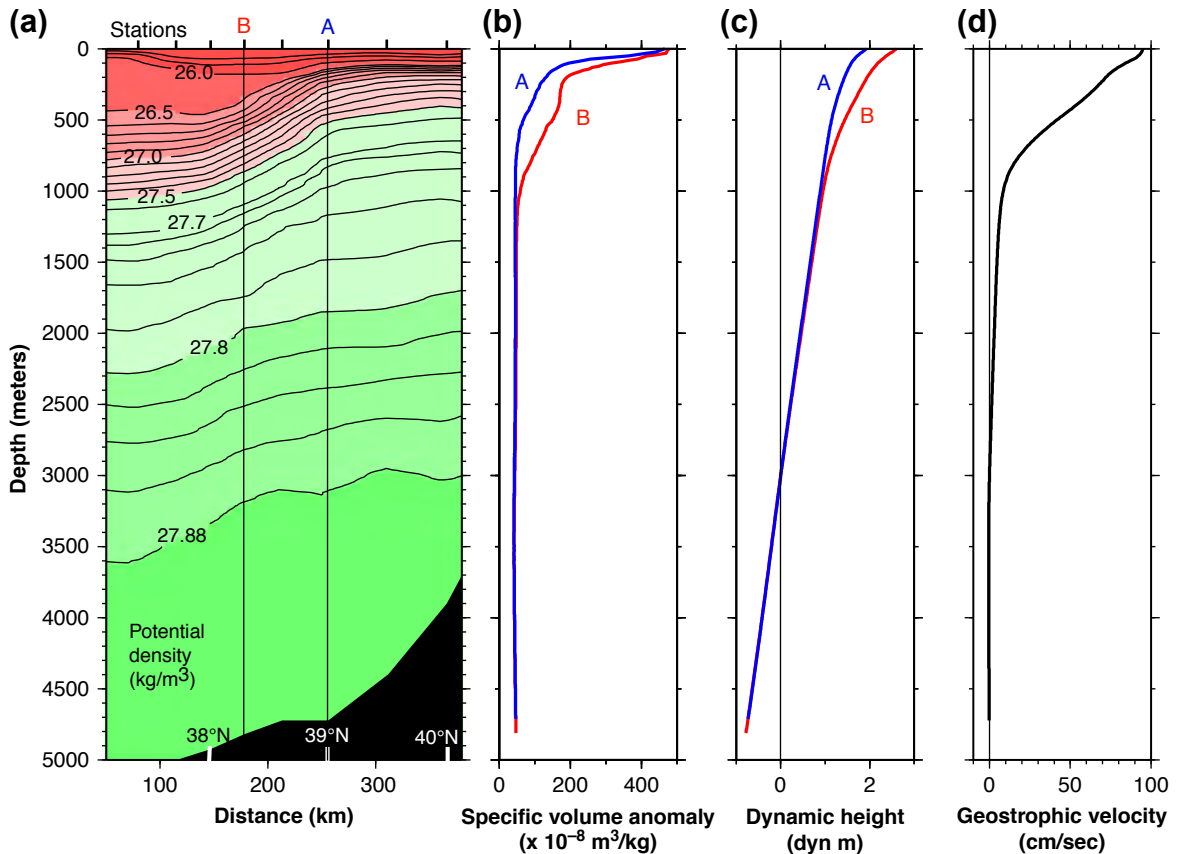


FIGURE 7.11 Geostrophic flow using observations. (a) Potential density section across the Gulf Stream (66°W in 1997). (b) Specific volume anomaly δ ($\times 10^{-8} \text{ m}^3/\text{kg}$) at stations A and B. (c) Dynamic height (dyn m) profiles at stations A and B, integrated from 3000 m depth. (d) Eastward geostrophic velocity (cm/sec), assuming zero velocity at 3000 m. This figure is described in detail in Section S7.6.2 of the online supplement.

7.6.3. Dynamic Topography and Sea Surface Height Maps

Dynamic height at one surface relative to another is the streamfunction for the geostrophic flow at that surface relative to the other, as an extension of Eq. (7.23f). Flows are along the contours with the high “hills” to the right of the flow in the Northern Hemisphere (to the left in the Southern Hemisphere). The speed at any point is proportional to the steepness of the slope at that point, in other words, inversely proportional to the separation of the contours.

Dynamic topography maps (equivalently, steric height) are shown in Chapter 14 and throughout the ocean basin chapters (9–13) to depict the geostrophic flow field. At the sea surface, all five ocean basins have highest dynamic topography in the west in the subtropics. The anticyclonic flows around these highs are called the *subtropical gyres*. The Northern Hemisphere oceans have low dynamic topography around 50° to 60°N; the cyclonic flows around these lows are the *subpolar gyres*. Tightly spaced contours along

the western boundaries indicate the swift western boundary currents for each of the gyres. Low values are found all the way around Antarctica; the band of tightly spaced contours to its north marks the eastward Antarctic Circumpolar Current. The contrast in dynamic height and sea-surface height from high to low in a given gyre is about 0.5 to 1 dynamic meters.

7.6.4. A Two-Layer Ocean

It is frequently convenient to think of the ocean as composed of two layers in the vertical, with upper layer of density ρ_1 and lower layer of density ρ_2 (Figure S7.21). The lower layer is assumed to be infinitely deep. The upper layer thickness is $h + H$, where h is the varying height of the layer above the sea level surface and H is the varying depth of the bottom of the layer. We sample the layers with stations at “A” and “B.” Using the hydrostatic equation (7.23c), we compute the pressure at a depth Z at the stations:

$$p_A = \rho_1 g(h + H) + \rho_2 g(Z - H) \quad (7.30a)$$

$$p_B = \rho_1 g(h_B + H_B) + \rho_2 g(Z - H_B) \quad (7.30b)$$

Here Z represents a common depth for both stations, taken well below the interface. If we assume that $p_A = p_B$, which amounts to assuming a “level of no motion” at Z , we can compute a surface slope, which we cannot measure in terms of the observed density interface slope:

$$\frac{h_A - h_B}{\Delta x} = \frac{\rho_2 - \rho_1}{\rho_1} \frac{H_A - H_B}{\Delta x} \quad (7.31a)$$

We then use Eq. (7.30a) to estimate the surface velocity v :

$$fv = g \frac{h_A - h_B}{\Delta x} = g \frac{\rho_2 - \rho_1}{\rho_1} \frac{H_A - H_B}{\Delta x} \quad (7.31b)$$

7.7. VORTICITY, POTENTIAL VORTICITY, ROSSBY AND KELVIN WAVES, AND INSTABILITIES

An apparent “problem” with the geostrophic balance (Eq. 7.23a,–b) is that it does not include any of the external forces that make the ocean flow; it has only pressure gradient and Coriolis force. How do we insert external forces such as the wind? In formal geophysical fluid dynamics, we would show that these forces *are* in the momentum equations, but are so weak that we safely consider the flows to be geostrophic (to lowest order). To reinsert the external forces, we have to consider the “vorticity” equation, which is formally derived from the momentum equations by combining the equations in a way that eliminates the pressure gradient force terms. (It is straightforward to do.) The resulting equation gives the time change of the vorticity, rather than the velocities. It also includes dissipation, variation in Coriolis parameter with latitude, and vertical velocities, which can be set externally by Ekman pumping.

The text that follows in this section is a greatly truncated version of the full text found at the textbook Web site (Section S7.7), which includes numerous figures and examples. For a more thorough treatment, it is recommended that the full text be used.

7.7.1. Vorticity

Vorticity is twice the angular velocity at a point in a fluid. It is easiest to visualize by thinking of a small paddle wheel immersed in the fluid (Figure 7.12). If the fluid flow turns the paddle wheel, then it has vorticity. Vorticity is a vector, and points out of the plane in which the fluid turns. The *sign* of the vorticity is given by the “right-hand” rule. If you curl the fingers on your right hand in the direction of the turning paddle wheel and your thumb points upward, then the vorticity is positive. If your thumb points downward, the vorticity is negative.

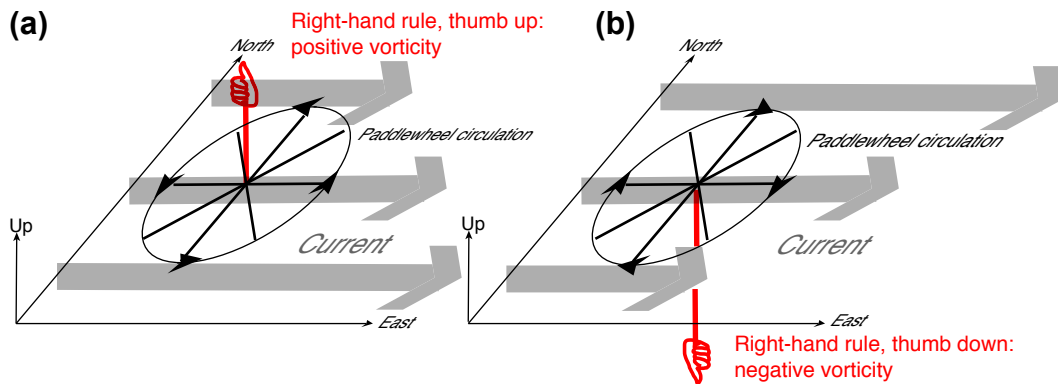


FIGURE 7.12 Vorticity. (a) Positive and (b) negative vorticity. The (right) hand shows the direction of the vorticity by the direction of the thumb (upward for positive, downward for negative).

The vorticity vector $\boldsymbol{\omega}$ is the curl of the velocity vector $\mathbf{v}$:

$$\begin{aligned}\boldsymbol{\omega} = \nabla \times \mathbf{v} = & \mathbf{i}(\partial v / \partial z - \partial w / \partial y) \\ & + \mathbf{j}(\partial w / \partial x - \partial u / \partial z) + \mathbf{k}(\partial u / \partial y - \partial v / \partial x)\end{aligned}\quad (7.32)$$

Vorticity, therefore, has units of inverse time, $(\text{sec})^{-1}$.

Fluids (and all objects) have vorticity simply because of Earth's rotation. This is called *planetary vorticity*. The vector planetary vorticity points upward, parallel to the rotation axis of Earth, which has an angular rotation rate of Ω :

$$\boldsymbol{\omega}_{\text{planetary}} = 2\boldsymbol{\Omega} \quad (7.33)$$

where $\Omega = 2\pi/\text{day} = 2\pi/86160 \text{ sec} = 7.293 \times 10^{-5} \text{ sec}^{-1}$, so $\omega_{\text{planetary}} = 1.4586 \times 10^{-4} \text{ sec}^{-1}$.

The vorticity of the fluid motion relative to Earth's surface (Eq. 7.32) is called the *relative vorticity*. The total or absolute vorticity of a piece of fluid is the sum of the relative vorticity and planetary vorticity.

For large-scale oceanography, only the local vertical component of the total vorticity is used because the fluid layers are thin compared with Earth's radius, so flows are nearly horizontal. The local vertical component of the

planetary vorticity is exactly equal to the Coriolis parameter f (Eq. 7.8c) and is therefore maximum and positive at the North Pole ($\phi = 90^\circ\text{N}$), maximum and negative at the South Pole ($\phi = 90^\circ\text{S}$), and 0 at the equator.

The local vertical component of the relative vorticity from Eq. (7.32) is

$$\zeta = (\partial v / \partial x - \partial u / \partial y) = \text{curl}_z \mathbf{v} \quad (7.34)$$

where $\mathbf{v}$ is the horizontal velocity vector. The local vertical component of the absolute vorticity is therefore $(\zeta + f)$. The geostrophic velocities calculated from Eq. (7.23) (Section 7.6) are often used to calculate relative vorticity.

7.7.2. Potential Vorticity

Potential vorticity is a dynamically important quantity related to relative and planetary vorticity. Conservation of potential vorticity is one of the most important concepts in geophysical fluid dynamics, just as conservation of angular momentum is a central concept in solid body mechanics. Potential vorticity takes into account the height of a water column as well as its local spin (vorticity). If a column is shortened and flattened (preserving mass), then it must spin more slowly. On the other hand, if a column is stretched and thinned (preserving

mass), it should spin more quickly similar to a spinning ice skater or diver who spreads his or her arms out and spins more slowly (due to conservation of angular momentum). Potential vorticity, when considering only the local vertical components, is

$$Q = (\zeta + f)/H \quad (7.35)$$

where H is the depth, if the fluid is unstratified. When the fluid is stratified, the equivalent version of potential vorticity is

$$Q = -(\zeta + f)(1/\rho)(\partial\rho/\partial z). \quad (7.36)$$

When there are no forces (other than gravity) on the fluid and no buoyancy sources that can change density, potential vorticity Q is conserved:

$$DQ/Dt = 0 \quad (7.37)$$

where “ D/Dt ” is the substantial derivative (Eq. 7.4). Figures S7.24–S7.26 and text describing the trade-offs between the relative, planetary, and stretching vorticity are found on the textbook Web site. All that we note here is that f varies with latitude, with huge consequences for ocean currents and stratification. Therefore, a special symbol is introduced to denote the change in Coriolis parameter with northward distance y , or in terms of latitude ϕ and Earth’s radius R_e :

$$\beta = df/dy = 2\Omega \cos \phi / R_e \quad (7.38)$$

We often refer to the “ β -effect” when talking about how changes in latitude affect currents, or the very large-scale, mainly horizontal Rossby waves for which the β -effect is the restoring force, described next.

7.7.3. Rossby Waves

The adjustment of any fluid to a change in forcing takes the form of waves that move out and leave behind a steady flow associated with the new forcing. We describe some general

properties of waves in Chapter 8. The large-scale, almost geostrophic circulation adjusts to changing winds and buoyancy forcing mainly through “planetary” or *Rossby waves* and *Kelvin waves* (Section 7.7.6). Pure Rossby (and Kelvin) waves are never found except in simplified models and lab experiments. However, much of the ocean’s variability can be understood in terms of Rossby wave properties, particularly the tendency for westward propagation relative to the mean flow.

Most of the physical motivation for these waves, including illustrations (Figures S7.27–S7.29), are in the full online version located at the textbook Web site. Only the most basic facts are in the following list.

1. Rossby waves have wavelengths of tens to thousands of kilometers. Therefore particle motions in Rossby waves are almost completely transverse (horizontal, parallel to Earth’s surface).
2. The restoring force for Rossby waves is the variation in Coriolis parameter f with latitude, so all dispersion information includes β (Eq. 7.38). As a water column is shoved off to a new latitude, its potential vorticity must be conserved (Eq. 7.37). Therefore, the water column height (long Rossby waves) or relative vorticity (short Rossby waves) begins to change. As with all waves, the column overshoots, and then has to be restored again, creating the wave.
3. All Rossby wave crests and troughs move only westward (relative to any mean flow, which could advect them to the east) in *both* the Northern and Southern Hemispheres; that is, the phase velocity is westward (plus a northward or southward component).
4. The group velocity of Rossby waves is westward for long wavelengths (more than about 50 km) and eastward for short wavelengths (even though the phase velocity is westward).

5. Velocities in Rossby waves are almost geostrophic. Therefore, they can be calculated from variations in pressure, for instance, as measured by a satellite altimeter, which observes the sea-surface height (e.g., Rossby wave-like behavior in Figures 14.18 and 14.19).

7.7.4. Rossby Deformation Radius and Rossby Wave Dispersion Relation

The length scale that separates long from short Rossby waves is called the *Rossby deformation radius*. It is the intrinsic horizontal length scale for geostrophic or nearly geostrophic flows, relative to which all length scales are compared. Again, see the full online version at <http://booksite.academicpress.com/DPO/> Chapter S7 for more information.

The Rossby deformation radius in an unstratified ocean is

$$R_E = (gH)^{1/2}/f \quad (7.39a)$$

where H is the ocean depth scale. R_E is called the *barotropic Rossby deformation radius* or “external” deformation radius. Barotropic deformation radii are on the order of thousands of kilometers.

The Rossby deformation radius associated with the ocean’s stratification is

$$R_I = NH_s/f \quad (7.39b)$$

where N is the Brunt-Väisälä frequency (7.15), and H_s is an intrinsic scale height for the flow. R_I is called the *baroclinic deformation radius* (or “internal” deformation radius). The vertical length scale H_s associated with the first baroclinic mode is about 1000 m, which is the typical pycnocline depth. R_I for the first baroclinic mode varies from more than 200 km in the tropics to around 10 km at high latitudes (Figure S7.28a) (Chelton et al., 1998).

The dispersion relation (Section 8.2) for first mode baroclinic Rossby waves is

$$\omega = \frac{-\beta k}{k^2 + l^2 + (1/R_I)^2} \quad (7.40)$$

where ω is the wave frequency, k and l are the wavenumbers in the east-west (x) and north-south (y) directions, β is as in Eq. (7.38), and R_I is as given in Eq. (7.39b). Highest frequency (shortest period) occurs at the wavelength associated with the Rossby deformation radius (Figure S7.29). The shortest periods vary from less than 50 days in the tropics to more than 2 to 3 years at high latitudes (Figure S7.28b from Wunsch, 2009). Poleward of about 40 to 45 degrees latitude there is no first baroclinic mode at the annual cycle, so seasonal atmospheric forcing cannot force the first baroclinic mode at these higher latitudes.

7.7.5. Instability of Geostrophic Ocean Currents

Almost all water flows are unsteady. When gyre-scale flows break up, they do so into large eddies, on the order of tens to hundreds of kilometers in diameter or larger (see Section 14.5). The size of the eddies is often on the order of the Rossby deformation radius. The eddies usually move westward, like Rossby waves.

Instabilities of flows are often studied by considering a mean flow and then finding the small perturbations that can grow exponentially. This approach is called “linear stability theory”; it is linear because the perturbation is always assumed to be small relative to the mean flow, which hardly changes at all. When perturbations are allowed to grow to maturity, when they might be interacting with each other and affecting the mean flow, the study has become nonlinear.

We define three states: *stable*, *neutrally stable*, and *unstable*. A stable flow returns to its original state after it is perturbed. A neutrally stable flow remains as is. In an unstable flow, the perturbation grows.

The two sources of energy for instabilities are the *kinetic energy* and the *potential energy* of the mean flow. Recall from basic physics that kinetic energy is $\frac{1}{2}mv^2$ where m is mass and v is speed; for a fluid we replace the mass with density ρ , or just look at the quantity $\frac{1}{2}\rho v^2$. Also recall from basic physics that potential energy comes from raising an object to a height; the work done in raising the object gives it its potential energy. In a stratified fluid like the ocean, there is no *available potential energy* if isopycnals are flat, which means that nothing has been moved and nothing can be released. For there to be usable or available potential energy, isopycnals must be tilted.

Barotropic instabilities feed on the kinetic energy in the horizontal shear of the flow. *Baroclinic instabilities* draw on the available potential energy of the flow. Baroclinic instability is peculiar to geostrophic flows, because Earth's rotation makes it possible to have a mean geostrophic flow with mean tilted isopycnals. On the other hand, barotropic instability is similar to instabilities of all sheared flows including those without Earth's rotation.

7.7.6. Kelvin Waves

Coastlines and the equator can support a special type of hybrid wave called a "Kelvin wave," which includes both gravity wave and Coriolis effects. Kelvin waves are "trapped" to the coastlines and trapped at the equator, which means that their amplitude is highest at the coast (or equator) and decays exponentially with offshore (or poleward) distance. Kelvin waves are of particular importance on eastern boundaries since they transfer information poleward from the equator. They are also central to how the equatorial ocean adjusts to changes in wind forcing, such as during an El Niño (Chapter 10).

Kelvin waves propagate with the coast to the right in the Northern Hemisphere and to the left in the Southern Hemisphere. At the equator, which acts like a boundary, Kelvin waves propagate only eastward. In their alongshore

direction of propagation, Kelvin waves behave just like surface gravity waves and obey the gravity wave dispersion relation (Section 8.3). However, unlike surface gravity waves, Kelvin waves can propagate in only one direction. Kelvin wave wavelengths are also very long, on the order of tens to thousands of kilometers, compared with the usual surface gravity waves at the beach. Although the wave propagation speed is high, it can take days to weeks to see the transition from a Kelvin wave crest to a Kelvin wave trough at a given observation point.

In the across-shore direction, Kelvin waves differ entirely from surface gravity waves. Their amplitude is largest at the coast. The offshore decay scale is the Rossby deformation radius (Section 7.7.4).

Lastly, Kelvin wave water velocities in the direction perpendicular to the coast are exactly zero. The water velocities are therefore exactly parallel to the coast. Moreover, the alongshore velocities are geostrophic, so they are associated with pressure differences (pressure gradient force) in the across-shore direction.

7.8. WIND-DRIVEN CIRCULATION: SVERDRUP BALANCE AND WESTERN BOUNDARY CURRENTS

The large-scale circulation in the ocean basins is asymmetric, with swift, narrow currents along the western boundaries, and much gentler flow within the vast interior, away from the side boundaries. This asymmetry is known as *westward intensification* of the circulation; it occurs in both the Northern and Southern Hemispheres and in the subtropical and subpolar gyres. Sverdrup (1947) first explained the mid-ocean vorticity balance now called the "Sverdrup interior" solution. Stommel (1948) and Munk (1950) provided the first (frictional) explanations for the western boundary

currents, and Fofonoff (1954) showed how very different the circulation would be without friction.

7.8.1. Sverdrup Balance

The gentle interior flow of the (non-equatorial) oceans can be described in terms of its meridional (north-south) direction. In the subtropical gyres, the interior flow is toward the equator in both the Northern and Southern Hemispheres. In the subpolar gyres, the interior flow is poleward in both hemispheres. These interior flow directions can be understood through a potential vorticity argument introduced by Sverdrup (1947), so we call the applicable physics the “Sverdrup balance.”

Consider a schematic of the subtropical North Pacific (Figure 7.13). The winds at the

sea surface are not spatially uniform (Figures 5.16a–c and S5.10a). South of about 30°N, the Pacific is dominated by easterly trade winds. North of this, it is dominated by the westerlies. This causes northward Ekman transport under the trade winds, and southward Ekman transport under the westerlies. As a result, there is Ekman convergence throughout the subtropical North Pacific (Figures 5.16d and S5.10a).

The convergent surface layer water in the subtropics must go somewhere so there is downward vertical velocity at the base of the (50 m thick) Ekman layer. At some level between the surface and ocean bottom, there is likely no vertical velocity. Therefore there is net “squashing” of the water columns in the subtropical region (also called *Ekman pumping*).

This squashing requires a decrease in either planetary or relative vorticity (Eq. 7.35). In the

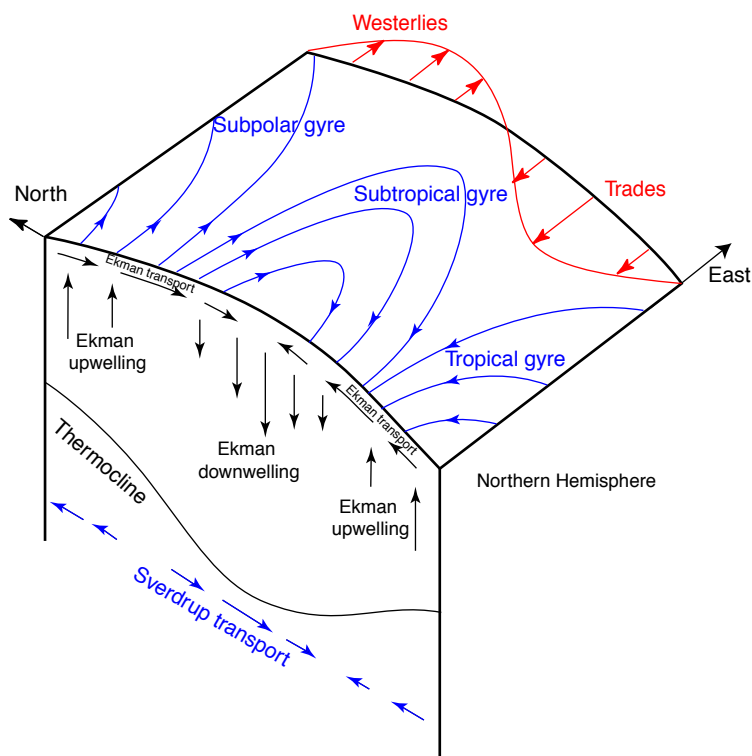


FIGURE 7.13 Sverdrup balance circulation (Northern Hemisphere). Westerly and trade winds force Ekman transport, creating Ekman pumping and suction and hence Sverdrup transport. See also Figure S7.12.

ocean interior, relative vorticity is small, so planetary vorticity must decrease, which results in the equatorward flow that characterizes the subtropical gyre (Figure S7.26).

The subpolar North Pacific lies north of the westerly wind maximum at about 40°N. Ekman transport is therefore southward, with a maximum at about 40°N and weaker at higher latitudes. Therefore there must be upwelling (*Ekman suction*) throughout the wide latitude band of the subpolar gyre. This upwelling stretches the water columns (Eq. 7.35), which then move poleward, creating the poleward flow of the subpolar gyre.

The *Sverdrup transport* is the net meridional transport diagnosed in both the subtropical and subpolar gyres, resulting from planetary vorticity changes that balance Ekman pumping or Ekman suction.

All of the meridional flow is returned in western boundary currents, for reasons described in the following sections. Therefore, subtropical gyres must be anticyclonic and subpolar gyres must be cyclonic.

Mathematically, the Sverdrup balance is derived from the geostrophic equations of motion with variable Coriolis parameter f (Eq. 7.23a,b). The x - and y -momentum equations are combined to form the vorticity equation:

$$f(\partial u/\partial x + \partial v/\partial y) + \beta v = 0 \quad (7.41)$$

Using the continuity equation

$$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0 \quad (7.42)$$

Eq. (7.41) becomes the potential vorticity balance

$$\beta v = f \partial w/\partial z. \quad (7.43)$$

This important equation states that water column stretching in the presence of rotation is balanced by a change in latitude (Figure S7.26).

In Eq. (7.43), the vertical velocity w is due to Ekman pumping. From Eqs. (7.20) and (7.21):

$$w = \partial/\partial x(\tau^{(y)}/\rho f) - \partial/\partial y(\tau^{(x)}/\rho f) = \text{"curl } \tau \text{"} \quad (7.44)$$

where τ is the vector wind stress, $\tau^{(x)}$ is the zonal wind stress, and $\tau^{(y)}$ is the meridional wind stress. Assuming that the vertical velocity w is zero at great depth, Eq. (7.43) can be vertically integrated to obtain the *Sverdrup balance*:

$$\begin{aligned} \beta(M^{(y)} - (\tau^{(x)}/f)) &= \partial/\partial x(\tau^{(y)}) - \partial/\partial y(\tau^{(x)}) \\ &= \text{"curl } \tau \text{"} \end{aligned} \quad (7.45)$$

where the meridional (south-north) mass transport $M^{(y)}$ is the vertical integral of the meridional velocity v times density ρ . The second term on the left side is the meridional Ekman transport. Thus, the meridional transport in the Sverdrup interior is proportional to the wind stress curl corrected for the Ekman transport.

The meridional transport $M^{(y)}$ is the *Sverdrup transport*. A global map of the Sverdrup transport integrated from the eastern to the western boundary is shown in Figure 5.17. The size of the integral at the western boundary gives the western boundary current transport since Sverdrup's model must be closed with a narrow boundary current that has at least one additional physical mechanism beyond those in the Sverdrup balance (a shift in latitude because of water column stretching driven by Ekman transport convergence). Physics of the boundary currents are discussed in the following sections.

7.8.2. Stommel's Solution: Westward Intensification and Western Boundary Currents

Because the Sverdrup balance applies to the whole ocean basin, the return flow must be in a narrow, swift meridional jet where the potential vorticity balance is different from the Sverdrup balance. Stommel (1948) included dissipation of potential vorticity Q on the right-hand side of Eq. (7.37), and showed that the returning flow must be along the western boundary (Figure S7.31). His potential vorticity balance is change in planetary vorticity

balanced by bottom friction. Stommel's idealized circulation resembles the western-intensified Gulf Stream and Kuroshio subtropical gyres. Much more discussion is provided in Section S7.8.2 in the textbook Web site <http://booksite.academicpress.com/DPO/>.

7.8.3. Munk's Solution: Western Boundary Currents

Like Stommel, Munk (1950) also showed western intensification of the gyres, but used a more realistic type of dissipation that could work equally well with a stratified ocean. To the potential vorticity conservation (Eq. 7.37), Munk added friction between the currents and the side walls. A narrow, swift jet along the western boundary returns the Sverdrup interior flow to its original latitude (Figure S7.32).

We use Munk's model to understand why the returning jet must be on the western rather than the eastern boundary (Figure 7.14). In the Sverdrup interior of a subtropical gyre, Ekman pumping squashes the water columns, which then move equatorward to lower planetary vorticity. To return to a higher latitude (i.e., increase the planetary vorticity), higher vorticity must be put back into the fluid through either stretching or relative vorticity. Stretching through wind stress curl in very narrow regions does not occur where and when it would be needed. Therefore, there must be an input of relative vorticity. Relative vorticity in a narrow boundary current is high because the horizontal shear is high. At the side wall, the velocity parallel to the wall is zero; it increases offshore, so the boundary current has positive relative vorticity if it is on a western boundary (Figure 7.14a). This vorticity is injected into the fluid by the friction at the wall, which then allows the planetary vorticity to change and the fluid to return to its original latitude; such a circulation closure would not be possible if the boundary current were on the eastern boundary (Figure 7.14b). The online version of this text

includes a more detailed explanation (see textbook Web site <http://booksite.academicpress.com/DPO/>).

7.8.4. Fofonoff's Solution: Large-Scale Inertial Currents

In addition to interior flows created from Ekman pumping or other external sources of vorticity, there are free, unforced modes of circulation, as shown by Fofonoff (1954), in a model with no wind input and no friction (Figure S7.34). This type of circulation is called an "inertial circulation". Without an external vorticity input, the interior flow is *exactly* zonal (east-west), because there is no way to change its planetary vorticity (due to the β -effect). Suppose there is westward flow across the middle of the ocean. When it reaches the western boundary, it gets back to the eastern boundary to feed back into the westward flow by moving either northward or southward along the western boundary in a very narrow current that can have as much relative vorticity as needed. Suppose it is northward. Then the relative vorticity of this frictionless current is positive, allowing it to move to a higher latitude. It then jets straight across the middle of the ocean, reaches the eastern boundary, and forms another narrow jet and moves southward, feeding into the westward flow in the interior.

Aspects of the Fofonoff inertial solution are found in highly energetic regions, such as near the separated Gulf Stream where its transport increases far above that predicted by the Sverdrup balance. The recirculation gyres associated with this energetic part of the Gulf Stream can be partially thought of in terms of Fofonoff gyres.

7.8.5. Wind-Driven Circulation in a Stratified Ocean

What happens to the wind-driven circulation theories in a stratified ocean? Water moves

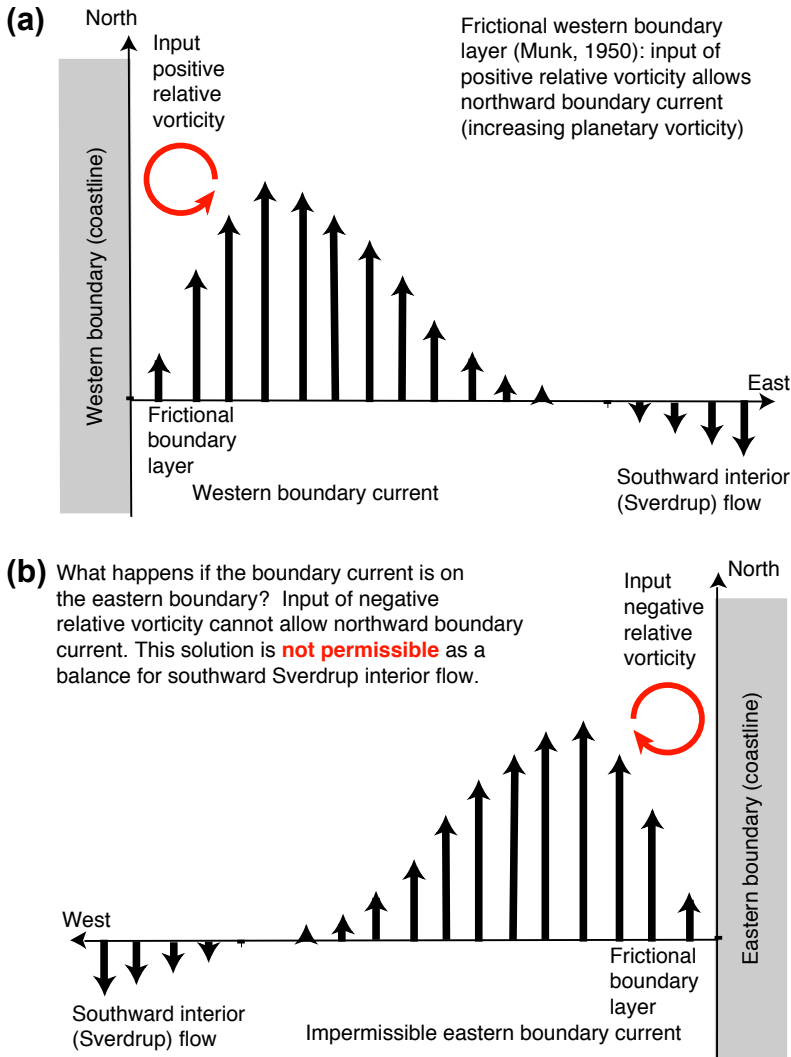


FIGURE 7.14 (a) Vorticity balance at a western boundary, with sidewall friction (Munk's model). (b) Hypothetical eastern boundary vorticity balance, showing that only western boundaries can input the positive relative vorticity required for the flow to move northward.

down into the ocean, mostly along very gradually sloping isopycnals. Where streamlines of flow are connected to the sea surface, we say the ocean is directly *ventilated* (Figure 7.15). Where there is Ekman pumping (negative wind stress curl), the Sverdrup interior flow is equatorward (Section 7.8.1). Water columns at the local mixed layer density move equatorward and encounter less dense water at the

surface. They slide down into the subsurface along isopycnals, still moving equatorward. This process is called *subduction* (Luyten, Pedlosky, & Stommel, 1983), using a term borrowed from plate tectonics. The subducted waters then flow around the gyre and enter the western boundary current if they do not first enter the tropical circulation. The details are beyond the scope of this text.

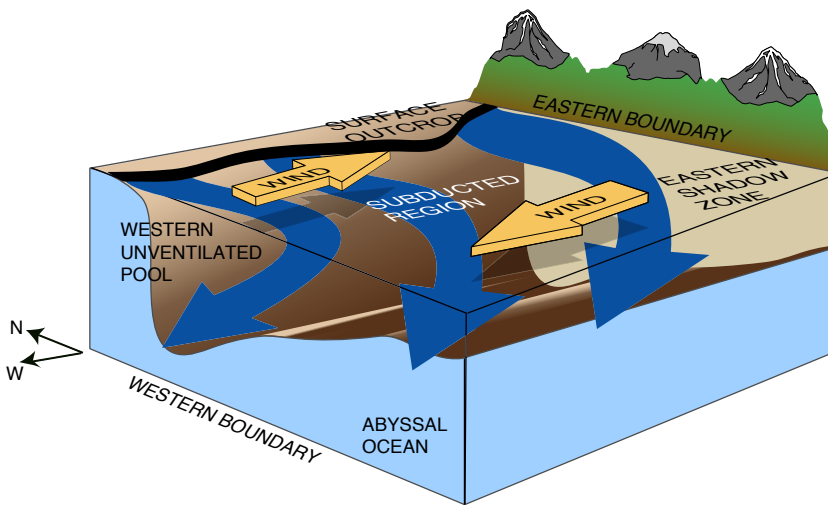


FIGURE 7.15 Subduction schematic (Northern Hemisphere). See Figure S7.35 for additional schematics, including obduction.

In each subducted layer, there can be three regions (Figure 7.15): (1) a *ventilated region* connected from the sea surface as just described, (2) a western unventilated pool with streamlines that enter and exit from the western boundary current without entering the surface layer, and (3) an eastern quiet (*shadow*) zone between the easternmost subducting streamline and the eastern boundary. A continuous range of surface densities is found in the subtropical gyre; the water column is directly ventilated over this full range, with waters at each density coming from a different seasurface location depending on the configuration of streamlines on that isopycnal. This is called the “*ventilated thermocline*”; in water mass terms, this process creates the Central Water. The maximum density of the ventilated thermocline is set by the maximum winter surface density in the subtropical gyre (Stommel, 1979).

The opposite of subduction is *obduction*, borrowed again from plate tectonics by Qiu and Huang (1995). In obducting regions, waters from subsurface isopycnals come up and into the surface layer. These are generally upwelling regions such as the cyclonic subpolar gyres and

the region within and south of the Antarctic Circumpolar Current.

Wind-driven circulation occurs in non-ventilated stratified regions as well. It is most vigorous in regions connected to the western boundary currents where water can enter and exit the western boundary. In these regions, the western boundary currents and their separated extensions usually reach to the ocean bottom. These dynamics are also beyond our scope.

7.9. WIND-DRIVEN CIRCULATION: EASTERN BOUNDARY CURRENTS AND EQUATORIAL CIRCULATION

7.9.1. Coastal Upwelling and Eastern Boundary Currents

The eastern boundary regions of the subtropical gyres have strong but shallow flow that is dynamically independent of the open ocean gyre regimes. Upper ocean eastern boundary circulation is driven by alongshore wind stress that creates onshore (or offshore) Ekman transport that creates upwelling (or

downwelling; Section 7.5.4). Beneath or inshore of the equatorward eastern boundary currents there is a poleward undercurrent or counter-current. Coastal upwelling systems are not restricted to eastern boundaries; the southern coast of the Arabian peninsula has the same kind of system. These circulations are fundamentally different from western boundary currents, which are tied to potential vorticity dynamics (Section 7.8).

The classical explanation of eastern boundary currents is that equatorward winds force Ekman flow offshore, which drives a shallow upwelling (on the order of 200 m deep) in a very narrow region adjacent to the coast (on the order 10 km; Figure 7.6c). The upwelling speed is about 5–10 m/day. Because of stratification, the source of upwelled water is restricted to layers close to the sea surface, usually between 50 and 300 m.

The zone of coastal upwelling can be extended to more than 100 km offshore by an increase in longshore wind strength with distance offshore; this is observed in each eastern boundary upwelling system due to topographic steering of the winds by the ocean-land boundary. The offshore Ekman transport therefore increases with distance offshore, which requires upwelling through the whole band (Bakun & Nelson, 1991). The zone is identified by positive wind stress curl (Figure 5.16d).

Upwelled water is cooler than the original surface water. It originates from just below the euphotic zone and therefore is also rich in nutrients. Cool surface temperatures and enhanced biological productivity are clear in satellite images.

Upwelling is strongly seasonal, due to seasonality in the winds. Onset of upwelling can be within days of arrival of upwelling-favorable winds. In one example, off the coast of Oregon, the surface temperature dropped by 6°C in two days after a longshore wind started.

Coastal upwelling is accompanied by a rise in upper ocean isopycnals toward the coast

(Figure 7.6). This creates an equatorward geostrophic surface flow, the *eastern boundary current*. These currents are narrow (<100 km width and near the coast), shallow (upper 100 m), strong (40 to 80 cm/sec), and strongly seasonal. The actual flow in an eastern boundary current system includes strong, meandering eddies and offshore jets/filaments of surface water, often associated with coastline features such as capes (Figure 10.6). Actual eastern boundary currents are some distance offshore, at the axis of the upwelling front created by the offshore Ekman transport.

Poleward undercurrents are observed at about 200 m depth beneath the equatorward surface currents. They are driven by a poleward pressure gradient force along the eastern boundary. When upwelling-favorable winds weaken or disappear, the equatorward flow also disappears and the poleward undercurrent extends up to the surface (there is no longer an undercurrent).

7.9.2. Near-Surface Equatorial Currents and Bjerknes Feedback

Circulation within about 2° latitude of the equator is very different from non-equatorial circulation because the Coriolis parameter f vanishes at the equator. Equatorial circulation is driven by easterly trade winds in the Pacific and Atlantic and by the seasonally reversing monsoonal winds in the Indian Ocean.

Since the Coriolis parameter vanishes and there is no frictional Ekman layer at the equator, the easterly trade winds drive equatorial surface flow due westward in a frictional surface layer (“normal panel” in Figure 10.27). The westward surface current is shallow (50 to 100 m) and of medium strength (10 to 20 cm/sec). In each of the three oceans, this westward surface flow is a part of the *South Equatorial Current*. The water piles up gently in the west (to about 0.5 m height) and leaves a depression in the east.

This creates an eastward pressure gradient force (from high pressure in the west to low pressure in the east). The pressure gradient force drives an eastward flow called the *Equatorial Undercurrent* (EUC). The EUC is centered at 100 to 200 m depth, just below the frictional surface layer. The EUC is only about 150 m thick. It is among the strongest ocean currents (>100 cm/sec). The pileup of waters in the western equatorial region also results in a deepened pycnocline there, called the *warm pool*, and a shoaling of the pycnocline in the eastern equatorial region (upwelling into the *cold tongue*). Coriolis effects set in a small distance from the equator, and off-equatorial Ekman transport enhances upwelling in the equatorial band. This enhances the cold tongue. The east-west contrast in temperature along the equator maintains the atmosphere's *Walker circulation*, which has ascending air over the warm pool and descending over the eastern colder area.

The Walker circulation is an important part of the trade winds that create the warm pool and cold tongue, so there can be a feedback between the ocean and atmosphere; this is called the *Bjerknes feedback* (Bjerknes, 1969) (Figure S7.36b located on the textbook Web site). If something weakens the trade winds, as happens at the beginning of an El Niño event (Chapter 10), the westward flow at the equator weakens and upwelling weakens or stops. Surface waters in the eastern regions therefore warm. Water in the deep warm pool in the west sloshes eastward along the equator, thinning the pool. The change in SST weakens the Walker circulation/trade winds even more, which further exacerbates the ocean changes. This is an example of a *positive feedback*.

In the Indian Ocean, the prevailing equatorial winds are monsoonal, meaning that trade winds are only present for part of the year. This creates seasonally reversing equatorial currents and inhibits the formation of the warm pool/cold tongue structure. The Indian Ocean sea surface temperature is high at all longitudes.

7.10. BUOYANCY (THERMOHALINE) FORCING AND ABYSSAL CIRCULATION

Heating and cooling change the ocean's temperature distribution, while evaporation, precipitation, runoff, and ice formation change the ocean's salinity distribution (Chapters 4 and 5). Collectively, these are referred to as *buoyancy*, or *thermohaline*, forcing. Buoyancy processes are responsible for developing the ocean's stratification, including its abyssal properties, pycnocline, thermocline, halocline, and upper layer structure (other than in wind-stirred mixed layers).

Abyssal circulation refers to the general category of currents in the deep ocean. The *overturning circulation*, also called the *thermohaline circulation*, is the part of the circulation associated with buoyancy changes, and overlaps spatially with the wind-driven upper ocean circulation; it also includes shallow elements that are independent of the abyssal circulation. In the overturning circulation, cooling and/or salinification at the sea surface causes water to sink. This water must rise back to the warm surface, which requires diffusion of heat (buoyancy) downward from the sea surface. The source of eddy diffusion is primarily wind and tidal energy.

7.10.1. Buoyancy Loss Processes (Diapycnal Downwelling)

Water becomes denser through net cooling, net evaporation, and brine rejection during sea ice formation. We have already described brine rejection (Section 3.9); it is responsible for creating the densest bottom waters in the global ocean (Antarctic Bottom Water and parts of the Circumpolar Deep Water) and also in the regional basins where it is operative (Arctic Ocean, Japan Sea, etc.). Here we focus on *convection* created by net buoyancy loss in the open ocean, when surface water becomes denser

than water below, and advects and mixes downward. Diurnal (daily) convection occurs at night in areas where the surface layer restratifies strongly during the day. During the annual cycle, cooling usually starts around the autumnal equinox and continues almost until the spring equinox. The resulting convection eats down into the surface layer, reaching maximum depth and density at the end of winter when the cumulative cooling reaches its maximum. A convective mixed layer can be hundreds of meters thick by the end of winter, whereas a wind-stirred mixed layer is limited to about 150 m by the depth of wind-driven turbulence.

Ocean convection is usually driven by surface cooling. Excess evaporation can also create convection, but the latent heat loss associated with evaporation is usually dominant. “Deep” convection is a loose term that usually refers to creation of a surface mixed layer that is thicker than about 1000 m. Deep convection has three phases: (1) preconditioning (reduction in stratification), (2) convection (violent mixing), and (3) sinking and spreading. (See Killworth, 1983 and Marshall & Schott, 1999.)

Convective regions have a typical structure (Figure S7.37). These include: (a) a *chimney*, which is a patch of 10 km to more than 100 km across within which preconditioning can allow convection and (b) convective *plumes* that are about 1 km or less across. The plumes are about the same size across as they are deep.

Deep convection occurs in only a very few special locations around the world: Greenland Sea, Labrador Sea, Mediterranean Sea, Weddell Sea, Ross Sea, and Japan (or East) Sea. These sites, with the exception of the isolated Japan Sea, ventilate most of the deep waters of the global ocean.

7.10.2. Diapycnal Upwelling (Buoyancy Gain)

The structure of the basin and global scale overturning circulations depends on both the

amount of density increase in the convective source regions and the existence of a buoyancy (heat) source at lower latitudes that is at least as deep as the extent of the cooling (Sandström, 1908; Figure S7.38). Since there are no significant local deep heat sources in the world ocean, waters that fill the deep ocean can only return to the sea surface as a result of diapycnal eddy diffusion of buoyancy (heat and freshwater) downward from the sea surface (Sections 7.3.2 and 5.1.3).

Munk’s (1966) diapycnal eddy diffusivity estimate of $\kappa_v = 1 \times 10^{-4} \text{ m}^2/\text{sec}$ (Section 7.3) was based on the idea of isolated sources of deep water and widespread diffusive upwelling of this deep water back to the surface. From all of the terms in the temperature and salt equations (7.12) and (7.13), Munk assumed that most of the ocean is dominated by the balance

$$\text{vertical advection} = \text{vertical diffusion} \quad (7.46a)$$

$$w \partial T / \partial z = \partial / \partial z (\kappa_v \partial T / \partial z) \quad (7.46b)$$

Munk obtained his diffusivity estimate from an average temperature profile and an estimate of about 1 cm/day for the upwelling velocity w , which can be based on deep water formation rates and an assumption of upwelling over the whole ocean. The observed diapycnal eddy diffusivity in the open ocean away from boundaries is much smaller than Munk’s estimate, which must be valid for the globally averaged ocean structure. This means that there must be much larger diffusivity in some regions of the ocean, now thought to be at the boundaries, at large seamount and island chains, and possibly the equator (Section 7.3).

7.10.3. Stommel and Arons’ Solution: Abyssal Circulation and Deep Western Boundary Currents

Deep ocean circulation has been explained using potential vorticity concepts that are very familiar from Sverdrup balance (Section 7.8;

Stommel, 1958; Stommel & Arons, 1960a,b). The sources of deep water are very localized. The deep water fills the deep ocean layer, which would raise the upper interface of this layer if there were no downward eddy diffusion (Section 7.10.2). This upwelling stretches the deep ocean water columns. Stretching requires a poleward shift of the water columns to conserve potential vorticity (Eq. 7.36). The predicted interior flow is therefore counterintuitive — it runs toward the deepwater source! (Actual abyssal flow is strongly modified from this by the major topography that modifies the β -effect, by allowing stretched columns to move toward shallower depths rather than toward higher latitude.)

Deep Western Boundary Currents (DWBCs) connect the isolated deepwater sources and the interior poleward flows. Whereas unambiguous poleward flow is not observed in the deep ocean interior (possibly mostly because of topography), DWBCs are found where they are predicted to occur by the Stommel and Arons abyssal circulation theory (Figure 7.16; Warren, 1981). One such DWBC runs southward beneath the Gulf Stream, carrying dense waters from the Nordic Seas and Labrador Sea. Swallow and Worthington (1961) found this

current after being convinced by Stommel to go search for it.

7.10.4. Thermohaline Oscillators: Stommel's Solution

An entirely different approach to understanding the meridional overturning circulation considers changes in overturn due to changes in dense water production by reducing the ocean to just a few boxes (Stommel, 1961). Such box models show how even the simplest model of climate change, for example, can lead to complex results. In this case, *multiple equilibria* result, that is, the system can jump suddenly between quite different equilibrium states. Stommel reduced the ocean to two connected boxes representing dense, cold, fresh high latitudes and light, warm, saltier low latitudes (Figure S7.41). In each box, the temperature and salinity are set by: (1) flux of water between the boxes (thermohaline circulation) that depends on the density difference between the boxes and (2) restoring temperature and salinity to a basic state over some set time period.

Stommel found that several different thermohaline circulation strengths exist for a given set

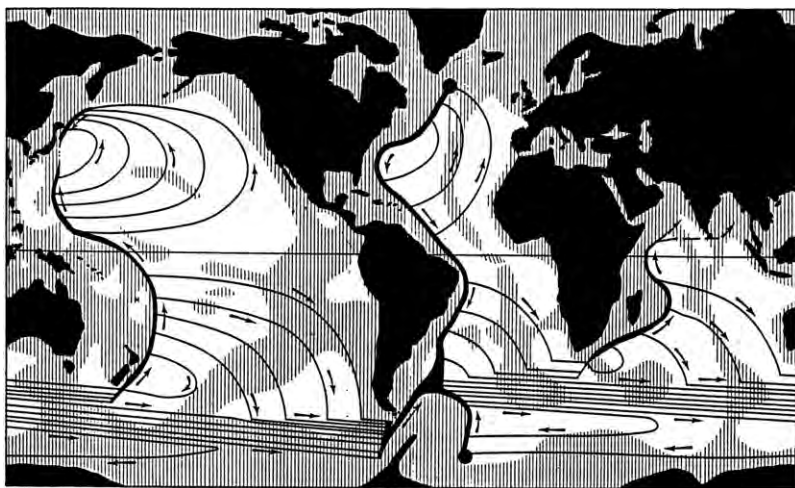


FIGURE 7.16 Global abyssal circulation model, assuming two deep water sources near Greenland and Antarctica (filled circles), filling a single abyssal layer. (These sources are actually at different densities.) Source: From Stommel (1958).

of choices of model parameters (externally imposed temperature and salinity, restoration timescales for temperature and salinity, and factor relating the flow rate to the density difference between the boxes). As the basic state was slowly changed, perhaps by reduction of the basic high-latitude salinity (which reduces its density), the flow rate slowly changed and then suddenly jumped to a different equilibrium rate. When the basic state salinity was then slowly increased, the system jumped back to a higher flow rate but at a very different basic salinity than during its decreasing phase. Thus this system exhibits *hysteresis*: it has different

equilibrium states depending on whether the state is approached from a much higher salinity or a much lower salinity.

The coupled atmosphere-sea-ice-land-physics-biology-chemistry climate system is far more complex than the two simple boxes in this very simple Stommel oscillator model. Yet its multiple equilibria and hysteresis behavior have been useful in demonstrating the potential for abrupt and relatively large changes in climate and, more specifically, for interpretation of numerical models of the changes in overturning circulation that could result from changes in external forcing.